MAX1000-FORTH

User Manual — a 16-bit Forth CPU on the Trenz MAX1000 (Intel MAX 10)

MAX1000-FORTH 1.3.1

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MAX1000-FORTH is a small interactive Forth system for the Trenz **MAX1000** (TEI0001-02, Intel MAX 10 10M08SAU169C8G). Its CPU, the **HF16**, is a 16-bit stack machine built in the FPGA. Every HF16 instruction is a Forth primitive. There is no soft processor and no second processor: the Forth kernel is part of the FPGA configuration, and Forth talks directly over the board's USB serial port, where the **MAX1000-FORTH app** or any terminal program talks to it.

Besides the board's 8 LEDs and its user button, Forth reaches the 23 digital pins of the Arduino-style header and the Pmod header, with four PWM channels, a DDS square-wave generator, two UARTs, two SPI masters and an I2C master on any of them, the 9 analog inputs and the temperature diode of the MAX 10's built-in converter, the board's LIS3DH **accelerometer**, and its 8 MB **serial flash**, which also keeps one Forth source that runs at every start.

This manual covers how to use the system, how the implementation works, and every word in the kernel. It describes MAX1000-FORTH 1.3.1 (firmware 1.3). For the hardware details (I/O map, pins, ADC channels, service requests), see design.md.

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1. Getting started

1.1 What you need

- A Trenz **MAX1000** (TEI0001-02, with the MAX 10 10M08SAU169C8G) and a micro-USB cable. MAX1000-FORTH supports this board only.
- **No programmer hardware:** the MAX1000 has its programmer on board (an Arrow USB Blaster in its USB chip), so the one USB cable writes the FPGA and carries Forth's serial port.
- On the PC, to write MAX1000-FORTH into the board once: the **Arrow USB Blaster** driver for that built-in programmer and Intel's **Quartus Prime Programmer** software, either the standalone Programmer or the one that comes with Quartus Prime.
- To work with it: on Windows the **MAX1000-FORTH app**, which comes with the release (chapter 2). Any system can use a terminal program instead.
- Optionally Python 3 with pyserial (`pip install pyserial`) for `tools/mforth.py`, which sends files and saves a source in the board's flash from the command line.

1.2 Put MAX1000-FORTH on the board

Paths in this manual are relative to the **MAX1000-FORTH folder**: the folder of an unpacked release, or the repository. Commands run in that folder.

1. Download `MAX1000-FORTH-1.3.zip` (or a later release) from the releases and unpack it. It holds:

Path	Content
`firmware\max1000_forth.pof`	the configuration for the MAX 10's internal flash
`firmware\max1000_forth.sof`	the same configuration for the FPGA only, until power-off
`firmware\program_flash.ps1`	writes one of the two to the board
`app\Max1000Forth.exe`	the MAX1000-FORTH app for Windows (chapter 2)
`tools\mforth.py`	sends lines and files, saves a source in the flash
`examples\`	Forth examples (`bench.fs`, section 6.8)
`docs\`	this manual (`Manual.pdf`) and the implementation details (`Implementation-Details.pdf`, `design.md`)
`README.md`, `LICENSE`, `CHANGELOG.md`	

2. Install the Arrow USB Blaster driver (for the board's built-in programmer) and the Quartus Prime Programmer software.
3. Plug the board in and run the script in PowerShell: `powershell .\firmware\program_flash.ps1` It finds `quartus_pgm` from Quartus Prime or the standalone Programmer and writes `firmware\max1000_forth.pof` into the MAX 10's internal configuration flash.
4. Press the board's **RESET** button (or unplug and plug the board in again). The MAX 10 loads the new configuration and Forth starts: **LED1** lights up.

The internal flash keeps MAX1000-FORTH until something else is written there; it replaces the configuration the board had before (for example the factory demo). The 8 MB serial flash is a separate chip, and the Forth source you save there (section 2.7) is not touched by programming the FPGA.

To try MAX1000-FORTH without replacing the board's configuration, `.\firmware\program_flash.ps1 -Sram` loads `firmware\max1000_forth.sof` into the FPGA only. It runs until the board loses power or RESET is pressed; then the configuration in the internal flash starts again.

From the repository, after building (chapter 8), the same is done with `quartus_pgm`:

```
quartus_pgm -c 1 -m JTAG -o "p;build/output_files/max1000_forth.pof"    # internal flash
quartus_pgm -c 1 -m JTAG -o "p;build/output_files/max1000_forth.sof"    # FPGA only
```

1.3 Connect

The MAX1000's USB chip (from FTDI) has two channels: channel A is the JTAG programmer, channel B the serial port Forth uses. Windows shows channel B as a USB serial port. Its number differs from PC to PC (COM3, COM5, COM17 ...); the Device Manager lists it under *Ports*, and this manual writes it as **COMx**. Start the app, choose that port and click **Connect** (chapter 2). Or use a terminal program:

Setting	Value
Port	the board's USB serial port, COMx
Speed	115200 baud, 8 data bits, no parity, 1 stop bit, no handshake
Enter key	sends CR (PuTTY's default)
Local echo	off (Forth echoes what you type)

Only one program can use the serial port at a time.

1.4 First session

```
HF16 Forth on MAX1000
2 3 + . 5  ok
: square dup * ;  ok
7 square . 49  ok
```

Type a line and press Enter. Forth runs the line from left to right and answers `ok`. Output appears on the same line as your input, as is usual in Forth. (To see the banner in a terminal program, connect first and then press Ctrl-R.)

Ctrl-R restarts Forth. It is handled in hardware, so it works at any time, also while a word is caught in an endless loop. It removes all words you have defined: the dictionary goes back to the kernel's. In the app, the **Restart** button does the same.

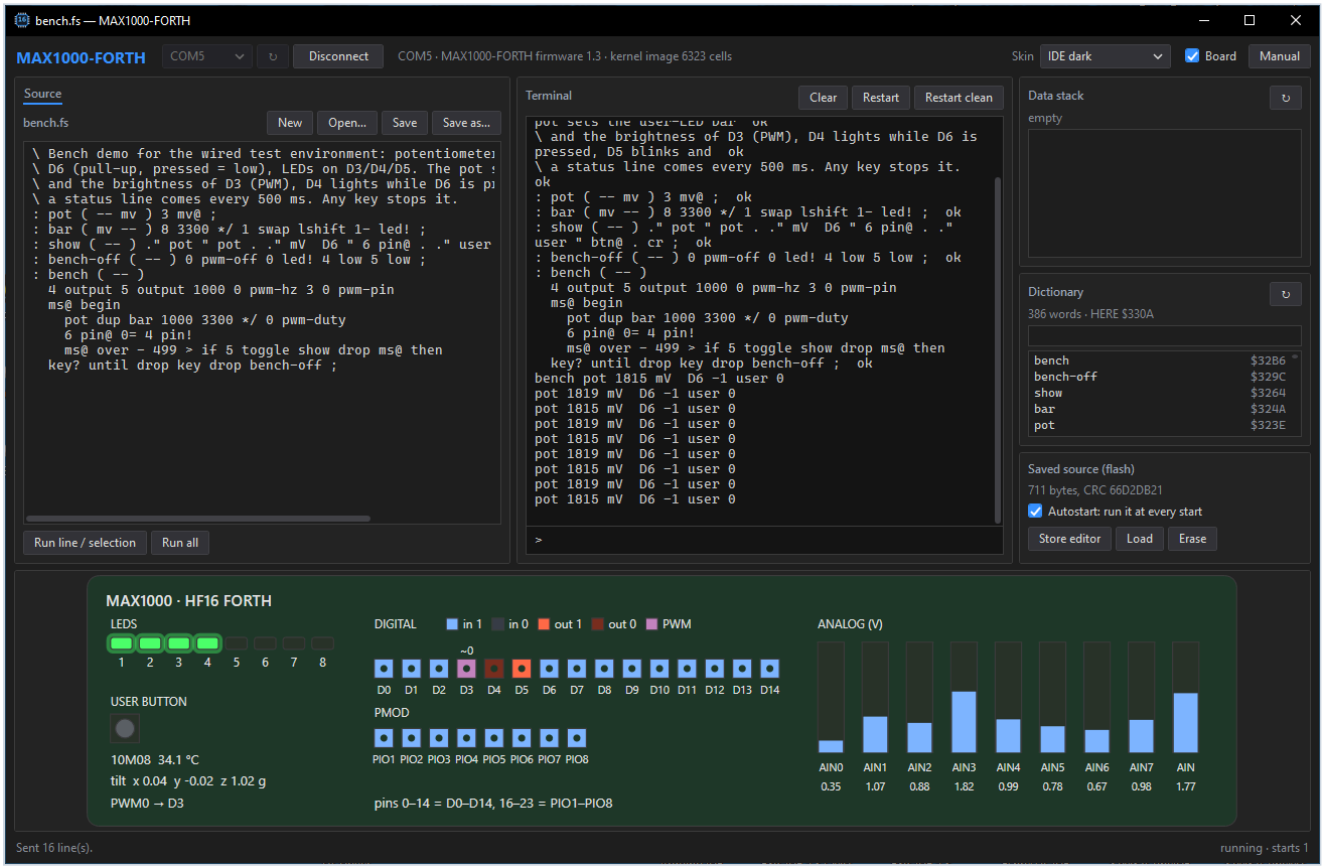
The board's **RESET** button is something else: it is the MAX 10's nCONFIG input and reloads the whole FPGA configuration from the internal flash, which also restarts Forth. Forth cannot read it. The **USER** button beside it belongs to Forth (`btn@`, section 5.10).

2. The MAX1000-FORTH app

The release contains the **MAX1000-FORTH app** for Windows (64-bit Windows 10 or 11): `app\Max1000Forth.exe`, a single program that needs no installation. Copy it anywhere and start it; a desktop shortcut to it is convenient. Its settings are kept in `%LocalAppData%\MAX1000-FORTH`, and its **Manual** button (F1) opens this manual.

The app is a small development environment for the HF16 on Windows:

- an **editor** for Forth source files, sent to Forth in whole or in part;
- a **terminal** that sends lines to Forth and shows what it prints;
- the **data stack** and the **dictionary**, read from the running Forth;
- the one **saved source** kept in the board's serial flash, which runs at every start;
- the **board view**: LEDs, user button, digital pins, PWM channels and analog inputs, live;
- five **skins**.



*The IDE dark skin after **Run all** of `examples/bench.fs` and `bench` (section 6.8): the potentiometer on AIN3 stands at about 1.8 V, so LED1 to LED4 are lit and PWM channel 0 drives D3; D4 is an output at 0, D5 blinks, and a status line comes every 500 ms.*

2.1 Connecting

1. Start **Max1000Forth.exe**.
2. Choose the board's port in the list at the top left (↻ looks for ports again) and click **Connect**. The app remembers the port.

3. The line beside the button shows the port, *MAX1000-FORTH firmware 1.3* and the size of the kernel image. The data stack, dictionary and saved-source boxes fill.

If the board does not answer, the app says so and disconnects: check that MAX1000-FORTH is on the board (chapter 1), that you chose channel B's port, and that no other program has the port open. **Disconnect** frees the port for a terminal program. While the app is connected, only it can use the port.

Start options, for a shortcut: `Max1000Forth.exe [--skin "Lab panel"] [--connect [COMx]] [file.fs]` connects at once (to COMx, or without it to the port the app remembers), chooses a skin, or opens a file in the editor.

2.2 The layout

Area	Place	What it is for
Top bar	top	Port, *Connect*, firmware; **Skin**, **Board** and **Manual** at the right
Editor	left	Forth source files, sent to Forth in whole or in part
Terminal	middle	Forth's output, the input line, the restarts
Data stack	top right	The stack at the last prompt, top first
Dictionary	right	The words Forth knows, newest first
Saved source (flash)	bottom right	The program in the board's flash and its autostart
Board view	below	The board's I/O, live (tick **Board**)
Status line	bottom	The last action's result (red for an error); at the right Forth's state and the starts since the configuration was loaded

The bars between the columns, and above the board view, can be dragged. **Skin** at the top right changes the look at once: *IDE dark*, *IDE light*, *Green phosphor*, *Amber* or *Lab panel* (which also switches the board view on).

2.3 How the app talks to the MAX1000

On the MAX1000 there is no second processor: the Forth kernel itself answers the app's requests (section 4.8). It answers them **at the prompt**, and whenever a program **waits in `ms`, polls `key?`, reads `key` or prints**. A word that only computes, such as a long `begin … until` loop without `ms` or output, gets no answers until it ends; meanwhile the status line shows *no answer* and the board view stands still until the word ends or waits.

Restart sends Ctrl-R, which the MAX1000 handles in hardware even during such a loop. **Restart clean** sends Ctrl-R and then ESC, which skips the saved source (section 4.7).

2.4 The terminal

- **Output** shows what Forth printed since the app connected, including the echo of each line it received. **Clear** empties it; Forth is not affected.
- **Input line** (the `>` at the bottom). Type a line and press **Enter**. **Up** and **Down** bring back earlier lines. The rules of section 3.1 apply: at most 80 characters per line, plain ASCII (other characters are sent as `?`).
- **Restart** restarts Forth (Ctrl-R). All words you defined are gone, and with autostart on, the saved source runs. **Restart clean** does the same without the saved source. Restart is also the way to stop a word that never returns, such as a `begin … again` loop or one waiting for `key`.

After each line the app reads the data stack; after a line that defines words (`:`, `create`, `variable`, `constant`, …) it also reads the dictionary.

2.5 The editor

- The title shows the file's name, or *untitled.fs*, and a star while there are unsaved changes; the window title repeats it.
- **New**, **Open…**, **Save** (Ctrl+S) and **Save as…** work on files, usually with the extension `.fs`. *New*, *Open…*, *Load* and closing the app ask before they discard unsaved changes.
- **Run line / selection**, or **F5** in the editor, sends the selected text. With nothing selected, it sends the line with the caret and moves the caret to the next line, so F5 steps through a file line by line.
- **Run all** (Ctrl+F5) sends the whole text to Forth, as if it were typed. The status line shows the progress; long texts go in pieces of 4 KB as Forth reads them, so nothing is lost. When Forth has read everything, the app reads the stack and the dictionary.

The text goes to Forth exactly as typed, so the rules of chapter 3 apply. A definition may span several lines. An error discards the rest of its line only (section 3.4): the following lines still run, so check the terminal after a long run. If an error interrupted a definition, fix it and run the definition again.

There is no *Pattern* tab with a MAX1000: the board has no pattern generator.

2.6 The data stack and the dictionary

Both are answers of the kernel; ↻ asks again.

Data stack

- It is the stack as Forth left it at the last `ok`, exactly what `.s` would print then, but top first: before each line, `quit` copies the stack into a buffer that the kernel sends to the app (`snap`, section 4.6). While a word runs, the box keeps showing the stack of the last prompt.
- Each item shows its position (0 is the top), its value in signed decimal and in hex. The line above gives the depth, or says the stack is empty.

Dictionary

- The line above the list gives the number of words and `HERE`, the next free byte (section 4.3).
- Each word shows its flags (`immediate`, `inline`, section 4.4) and the byte address of its code in hex. Your own words come first.
- The box above the list shows only the words containing its text.
- **Double-click** a word to add it to the terminal's input line.

2.7 The saved source

The serial flash keeps one Forth source. With **autostart** on, Forth reads it after the kernel starts, at power-up, after RESET and after every restart, as if it were typed (section 4.7).

- The line under the heading shows **none**, the length in bytes and the CRC, or **damaged** (its check failed: store it again or erase it).
- **Store editor** writes the editor's text to the flash; a source already stored is replaced only after you confirm. The app stores at most 65,520 bytes (`mforth.py` up to 65,535). Writing takes about one second per 10 KB, mostly the time the serial port needs. Storing does not run the source. It keeps the autostart setting of the

source it replaces; a first source is stored with autostart on. A line end is added after the last line if it has none.

- **Load** puts the saved source into the editor, for changes or a file of its own.
- **Erase** removes the saved source after you confirm.
- **Autostart: run it at every start** switches the autostart. It can be ticked only while a source is stored.

A saved source should end at the prompt. Its last line may start an application that runs until a key is pressed (section 6.9); one that loops for ever, or waits for `key`, holds Forth at every start. To get past it, **hold the USER button** while you press RESET or restart, or click **Restart clean**; then switch the autostart off or erase the source.

2.8 The board view

Tick **Board** at the top right (the Lab panel skin ticks it for you). The view shows the board as Forth sees it, read several times a second:

- the **LEDs** LED1..LED8 (`led!`) and the **USER** button, highlighted while pressed;
- the **digital pins** D0..D14 (Forth's pins 0..14) and, below them, PIO1..PIO8 on the Pmod header (pins 16..23). Each pin is coloured by what it does: an input reading 1 or 0, an output driving 1 or 0, or a pin owned by a PWM channel (marked `~` and the channel number). Open inputs have no pull-ups and may show either level (section 5.11);
- which PWM channels are on, and their pins (*PWM0 → D3*). A pin the DDS drives shows as an output (section 5.14), and so do the output pins of the serial interfaces (TX, SCK, MOSI; an I2C line only while it is pulled low). The app does not name the serial functions yet; the `board` answer carries them (section 4.8);
- the **analog inputs** AIN0..AIN7 and AIN as bars with their voltage, 0 to 3.3 V (section 5.12);
- the chip **temperature** and the **tilt** from the accelerometer in g (x, y, z) (sections 5.12 and 5.16).

The view only reads: Forth's I/O is never changed by it. It is updated only while Forth answers (section 2.3).

2.9 A worked session

This session writes a small program, tries it, and makes it start with the board. The program is a running light on LED1 to LED8:

```
\ lights.fs - a running light on LED1..LED8
: step ( n -- n' ) 2* dup 256 = if drop 1 then ;
: lights ( n -- ) 1 swap 0 do dup led! step 100 ms loop drop 0 led! ;
: hello ." lights ready" cr ;
hello
```

1. Click **New**, type the program into the editor, click **Save as...** and save it as `lights.fs`.
2. Click **Run all**. The terminal shows each line followed by `ok`, and the last line prints `lights ready`. `hello`, `lights` and `step` are now the first three words in the dictionary.
3. Tick **Board**. In the input line, type `16 lights` and press Enter. LED1 to LED8 light one after the other, twice, in 1.6 seconds, on the board and in the view. Press **Up** and Enter to run it again.
4. Type `1 2 3`: the data stack shows `3`, `2`, `1`, top first. Type `+` and it shows `5`, `1`. Type `drop drop`.
5. Click **Store editor**. The saved-source line shows the length and CRC, and **Autostart** is ticked.
6. Click **Restart**. Forth starts again, and the terminal shows the banner followed by `lights ready`: the saved source ran (without echo). From now on, `lights` is there every time the board starts.
7. To remove it, untick **Autostart** (the source stays in flash) or click **Erase**.

3. Working with the interpreter

3.1 Input line

- A line holds up to 80 characters; characters after the 80th are dropped (and not echoed).
- **Backspace** (or Delete) removes the last character.
- **Enter** runs the line. A CR or an LF ends a line, so a terminal that sends CR LF makes an extra empty line, which only prints another `ok`.
- After the line, Forth prints `ok` if it is interpreting. While you are in the middle of a colon definition (compiling), it prints no `ok`. A definition may span several lines.
- Characters typed while a word runs wait in the receive buffer (8 KB) until Forth reads them. Ctrl-R empties it.

3.2 Words and numbers

The input is split at spaces (control characters count as spaces). Each piece is looked up in the dictionary; if it is a word, it runs (or is compiled, inside a definition). If not, Forth tries to read it as a number. If that fails too, it is an error.

- **Names are case-insensitive:** `DUP`, `Dup` and `dup` are the same word.
- **Names** may contain any characters except space, up to 31 characters.
- **Numbers** are read in the current `base` (10 at start-up):

Input	Value
`42`	42
`-42`	−42
`$2A`	42 (hex regardless of `base`)
`-$2A`	−42
`2A` after `hex`	42
`115200.`	a double number: 49664 and 1 (115200 = 1 × 65536 + 49664), two cells
`$1C200.`	the same in hex

Numbers are 16 bits. Larger numbers wrap around silently: `70000 .` prints `4464` (70000 − 65536). A number with a **trailing dot** is an unsigned **double number**, two cells with the high cell on top (`ud`, 0..4,294,967,295); the baud rate of `uart-open` is one (section 5.15). A minus sign is not allowed there.

3.3 Stack notation

This manual describes each word's effect on the data stack as `( before -- after )`, with the top of the stack on the right. `( a b -- c )` means the word takes `b` (top) and `a`, and leaves `c`.

Symbol	Meaning
`n`	signed number (−32768..32767)

Symbol	Meaning
`u`	unsigned number (0..65535)
`x`	any cell
`f`	flag: 0 = false, −1 = true (any non-zero counts as true)
`c`	character or byte (0..255)
`a`, `addr`	byte address
`ud`	unsigned double number, two cells: low cell, then the high cell on top
`lo hi`	a 24-bit flash address: `lo` = bits 15..0, `hi` = bits 23..16 (section 5.17)
`xt`	execution token (address of a word's code)
`h`	header address of a dictionary entry
`"name"`	a name that follows in the input line

3.4 Errors

Message	Cause	What happens
`name ?`	word not found and not a number	data stack cleared, back to interpreting
`stack underflow`	a word took more items than were on the stack	data stack cleared
`stack overflow`	more than 20 items on the data stack	data stack cleared
`bad led`	`led-on`, `led-off`, `led-toggle` or `led-mask` got a number outside 1..8 (section 5.10)	data stack cleared
`bad pin`	a pin word got a number that is not a pin (section 5.11)	data stack cleared
`bad input`	`adc@` or `mv@` got an input above 8 (section 5.12)	data stack cleared
`bad channel`	a PWM word got a channel above 3 (section 5.13)	data stack cleared
`bad frequency`	`pwm-hz` got 0, `pwm-khz` a value outside 1..24000, `spi-khz` one outside 1..12000, `i2c-open` one outside 1..1000	data stack cleared
`pin taken`	a PWM channel, the DDS or a serial interface was to get a pin another one has (section 5.15)	data stack cleared, nothing changed
`bad uart`, `bad spi`	a UART number other than 0 and 1, an SPI number other than 0 and 1	data stack cleared
`bad baud`	`uart-open` got 0 or less than 3 Bd	data stack cleared
`bad bits`	`spi-bits` got a width other than 8 and 16	data stack cleared
`no ack`	the I2C device did not acknowledge in `i2c!` or `i2c@` (wrong address, no device, no pull-ups); a stop was sent	data stack cleared
`i2c timeout`	SCL stayed low for 22 ms (a device stretching too long, or no pull-up on SCL)	data stack cleared
`data RAM full`	`buffer:` asked for more than `unused-data` (section 5.7)	data stack cleared

After an error, Forth discards the rest of the line, clears the data stack and returns to interpreting. If the error happened inside a colon definition, that definition stays unfinished and hidden. It cannot be called, but it still occupies memory. If the error happened while the saved source was running, the message is followed by `(in the saved source)` and the rest of the saved source is skipped (section 4.7).

The stack check runs after each word typed at the terminal, not inside definitions. A word that pushes more than 28 items wraps the stack pointer around, and this is reported as underflow.

A few words report a problem without an error: `autostart` prints `no saved source`, and `store` `store timeout` or `store failed` (section 5.18). They leave the rest of the line running.

4. How the implementation works

4.1 The pieces

```
app / PuTTY ── USB ── FTDI channel B ── UART ──┐
                                               │
HF16 system in the MAX 10 ─────────────────────┘
  ├─ HF16 CPU, 48 MHz
  ├─ 24 KB RAM: 16 KB kernel + your words, 8 KB data
  ├─ 8 KB receive and 1 KB send FIFO
  ├─ 8 user LEDs, user button, ms timer
  ├─ 23 digital pins: D0..D14, PIO_01..PIO_08
  ├─ PWM: 4 channels on any pin
  ├─ DDS: square wave on any pin, 192 MHz clock
  ├─ 2 UARTs, 2 SPI masters, I2C master: any pin
  ├─ ADC: AIN, AIN0..AIN7, temperature diode
  ├─ SPI: serial flash (8 MB), LIS3DH accelerometer
  └─ CRC-32 unit
```

There is no ARM, no soft processor and no loader. The RAM's initial contents, the kernel image, are part of the FPGA configuration, so the kernel is in place the moment the MAX 10 has loaded its configuration. The serial port is connected straight to the CPU's FIFOs.

Part	File
HF16 CPU	`rtl/hf16_cpu.v`
Top level: PLL, RAM, UART, I/O registers, pins	`rtl/max1000_forth.v`
UART and its FIFOs	`rtl/uart.v`, `rtl/fifo_bram.v`
PWM	`rtl/pwm.v`
DDS square-wave generator	`rtl/dds.v`
UART, SPI master, I2C master for the pins	`rtl/uart_port.v`, `rtl/spi_port.v`, `rtl/i2c_port.v`
ADC scanner around Intel's Modular ADC core	`rtl/adc_scan.v`
SPI master (flash, accelerometer) and CRC-32 unit	`rtl/spi_flash.v`
Forth kernel, written in Forth	`forth/kernel.fs`
Cross-compiler and simulator	`tools/hf16.py`
Line sender, source storing	`tools/mforth.py`
Build scripts	`scripts/build.ps1`, `scripts/build.tcl`
The MAX1000-FORTH app (C#, WPF; chapter 2)	`app/`

4.2 The HF16 CPU

- **16-bit cells.** Numbers, addresses and flags are all 16 bits.
- **Two stacks in hardware**, 32 entries each. The **data stack** holds parameters. Its top (T) is a register, and the item below it (N) is directly available, so `+`, `swap` or `over` each take one clock. The **return stack** holds

return addresses, loop indices and items moved there with `>r`.

- **One instruction per clock** at 48 MHz (the board's 12 MHz oscillator times 4), so about 48 million primitives per second.
- **Five instruction types:** literal (push a number 0..32767), jump, conditional jump, call, and ALU. A single ALU instruction can also move the stack pointers, store to memory and return from a subroutine.
- **Fast calls and returns:** a call takes one clock. A return is usually merged into the last instruction of a word for free.
- Jumps and calls hold 13-bit cell addresses, so **code must lie in the first 16 KB** of the RAM. The 8 KB above are for data only.

The instruction encoding is described in the implementation details (`docs/design.md`, `Implementation-Details.pdf`, chapter 3).

4.3 Memory map

Address	Content
`$0000`	jump to `cold` (start-up)
`$0002`–`$3165`	the kernel: 376 words with their headers and variables
`$3166`–`$3FFF`	free for your definitions (3,738 bytes). `here` shows the next free byte, `unused` how much is left
`$4000`–`$5BFF`	data RAM for `buffer:` (7,168 bytes); `unused-data` shows how much is left
`$5C00`–`$5FFF`	the kernel's buffers (below; `$5E40`–`$5EFF` is not used)
`$6000`–`$7FFF`	no RAM; stores there are ignored
`$8000`–`$80FF`	I/O registers (below); they repeat every 256 bytes up to `$FFFF`

Cells are stored little-endian, low byte first. Cell addresses must be even: `@` and `!` ignore bit 0. There is no check for a full dictionary: watch `unused`. A definition that grows past `$3FFF` lands in the data RAM, where it cannot run, and overwrites the data of `buffer:`.

The kernel's buffers:

Address	Name	Use
`$5C00`–`$5CFF`	`pbuf`	one flash page while a source is stored
`$5D00`–`$5D51`	`tib`	the input line (82 bytes)
`$5D60`–`$5D9F`	`rq`	a service request from the app
`$5DA0`–`$5DDF`	`sbuf`	`.s` parks the stack here
`$5DE0`–`$5E07`	`ssnap`	the stack at the last prompt, for the app
`$5E10`–`$5E1F`	`hdr`	the saved source's header
`$5E20`–`$5E3F`	`pin-owner`	the owner of each pin (section 5.15)
`$5F00`–`$5FFF`	`lbuf`	the piece of saved source being interpreted

The I/O registers can also be read and written directly with `@` and `!`:

Address	Read	Write
`$8000`	next received byte (removes it from the receive FIFO)	send a byte
`$8002`	bit 0 = a byte has arrived, bit 1 = room to send, bit 2 = sending finished	–
`$8004`	LED1..LED8 (bits 0..7)	LED1..LED8
`$8006`	bit 0 = USER button pressed	–
`$8008`	–	any value: restart Forth, like Ctrl-R
`$800A`	millisecond counter	–
`$800C`	next received byte without removing it	–
`$800E`	number of bytes in the receive FIFO	–
`$8010`, `$8018`	port 0, 1: pin levels	–
`$8012`, `$801A`	port 0, 1: output latch	output latch
`$8014`, `$801C`	port 0, 1: direction, 1 = output	direction
`$8016`, `$801E`	port 0, 1: pins driven now (outputs and the pins of PWM, DDS and serial interfaces)	–
`$8020`	SPI: the byte received last	SPI: send a byte (starts an exchange)
`$8022`	bit 0 = flash selected, bit 1 = busy, bit 2 = accelerometer selected	bit 0 = select the flash, bit 1 = select the accelerometer
`$8024`	–	reset the CRC
`$8026`	–	add the low byte to the CRC
`$8028`, `$802A`	CRC-32, bits 15..0 and 31..16	–
`$8040`–`$8052`	ADC channels 0..8 and the temperature diode (slot 9), 12 bits	–
`$8060`–`$8064`	DDS: tuning low, high, control (section 5.14)	DDS registers
`$8080`–`$809E`	PWM channels 0..3 (section 5.13)	PWM registers
`$80A0`–`$80B8`	UART 0, 1 (section 5.15)	UART registers
`$80C0`–`$80DA`	SPI 0, 1 (section 5.15)	SPI registers
`$80E0`–`$80E4`	I2C (section 5.15)	I2C registers

Section 5.11 explains the ports, 5.12 the ADC, 5.13 the PWM registers, 5.14 the DDS, 5.15 the serial interfaces and 5.17 the flash SPI and CRC registers.

4.4 Dictionary entries

Every word has a header followed by its code:

```
h:      link     cell: address of the previous header (0 ends the list)
h+2:    count    byte: bits 0..4 = name length, bit 5 = inline, bit 6 = hidden,
                       bit 7 = immediate
h+3:    name     the name's characters, padded to an even address
xt:     code     HF16 instructions
```

`latest` holds the address of the newest header. Searching starts there and follows the links backwards, so a new definition with an existing name replaces the old one for everything compiled afterwards.

- **Immediate** words run even while compiling. Examples: `if`, `;`, `."`.
- **Hidden** marks a word whose definition isn't finished yet, so it can't be found.
- **Inline** words are copied into new definitions instead of being called. The primitives `dup`, `+`, `@` and others are single instructions. The return-stack words `>r`, `r>`, `r@`, `i` and `unloop` *must* be inline: as subroutines they would disturb their own return address.

4.5 What a colon definition compiles to

`: name ... ;` lays down a header and then, word by word:

In the source	Compiled as
a primitive (`dup`, `+`, `!`, `>r`, ...)	its instruction(s), copied inline
any other word	one call instruction
a number 0..32767	one literal instruction
any other number	a literal and an `invert` instruction
`if` / `while` / `until`	a conditional jump
`else` / `again` / `repeat`	a jump
`do`	`swap >r >r`, inline
`loop` / `+loop`	a call to `(loop)` / `(+loop)`, a conditional jump back, then `unloop`
`." text"`	a call to `(.")` followed by the text
`;`	a return instruction

Example: `: square dup * ;` compiles to three instructions: `dup`, `*`, return.

The cross-compiler that builds the kernel also merges a word's final return into the last instruction when that is free, and turns a final call into a jump. Definitions typed at the terminal don't get these two optimizations.

4.6 Start-up and restarts

Power-up or RESET. The MAX 10 loads its configuration from its internal flash, which also fills the RAM with the kernel image. When the PLL runs, the CPU starts at address 0, which jumps to `cold`.

Ctrl-R. The serial receiver catches the byte 0x12 (Forth never sees it) and resets the CPU, both serial FIFOs (typed-ahead input is discarded), the LEDs, the pins (all inputs again), the PWM channels, the DDS and the serial interfaces. The RAM keeps its contents, and the CPU starts at address 0 again. A program can do the same by writing any value to `$8008`. The millisecond counter keeps running.

`cold` then makes Forth fresh: it restores `here` and `latest` from `dp0` and `latest0`, so the dictionary is the kernel's again, frees the data RAM, counts the start, switches the accelerometer on (`accel-on`), switches the PWM channels, the DDS and the serial interfaces off and frees their pins (`units-off`), sets `base` to 10, prints the banner, switches LED1 on, runs the saved source if autostart is on (`autoload`, section 4.7) and enters `quit`, the interpreter loop. `quit` copies the data stack for the app's stack view (`snap`), reads a line (`accept-line`), interprets it (`interpret`), prints `ok`, and repeats.

Typing `cold` (or `0 execute`) runs the same word without the hardware reset: the dictionary is reset and the saved source runs again, but the pins keep their state (the PWM channels, the DDS and the serial interfaces are switched off).

4.7 The saved source (autostart)

One Forth source can be kept in the serial flash (from the app, section 2.7, or with `python tools/mforth.py -p COMx --store file.fs`). Its format:

Flash address	Content
`$000000` (a 4 KB sector)	header, 16 bytes, little-endian: magic `HF16`, length (32 bits), CRC-32 of the text, flags (bit 0 = autostart)
`$010000` (a 64 KB block)	the text, at most 65,535 bytes, lines end in CR

With autostart on, `cold` feeds it to Forth at power-up, after RESET and after every restart, exactly as if it were typed: the definitions are compiled and the commands run. Its lines are not echoed and get no `ok` (only the last one, which becomes the first prompt); only what they print appears. The words it defines are there when the first prompt comes, and its last line may start an application (section 6.9). The kernel itself is never changed, so a faulty source cannot break the start for good:

- **ESC within 200 ms** of the start skips it. The app's *Restart clean* and `mforth.py --skip` send Ctrl-R followed by ESC.
- **Hold the USER button** while you press RESET or restart, until `autostart skipped` appears (about a quarter of a second after the banner: the button is read after the wait for ESC). Forth prints `autostart skipped` in both cases.
- An **error** in a line of the source stops it: the message is followed by `(in the saved source)`, the words defined before the error exist, the rest of the source is not run.
- A source that never returns to the prompt (`begin again`, waiting for `key`) blocks all further input; restart clean, then fix or erase it.

Before it runs the source, `cold` checks the header and the CRC of the whole text; a damaged source does not run. `key` then takes its bytes from the saved source, which the kernel copies from the flash 256 bytes at a time into a buffer (`lbuf`). The flash is selected only while a piece is copied, so the lines of a saved source may use every word, including the flash and accelerometer words. Requests from the app are answered while the source runs, wherever it waits (`ms`, `key?`, output).

4.8 Service requests (the app)

The app speaks service protocol 1: a request is the byte SOH (0x01), a line of text and LF; the answer is `SOH ok <name> <length> LF <payload>` or `SOH err <name> <reason> LF`. The kernel answers it itself.

Requests arrive in the receive FIFO between terminal bytes. `service?` answers every request at the head of the FIFO; `key?` calls it and decides on the head byte itself, so a SOH is never taken as terminal input. `key`, `ms` and the wait in `emit` (while the send FIFO is full) go through `key?` or `service?`, which is why the app gets answers at the prompt and whenever a program waits, polls `key?` or prints. A request queued behind terminal input waits until Forth has read that input. A 0x01 in Forth's own output is sent twice, so the app can tell it from the start of an answer; a terminal program shows both.

Request	Answer
`version`	`firmware max1000-forth 1.3`, `protocol 1`, `kernel <cells> <checksum>`
`status`	state, receive FIFO bytes, starts, the saved source's state, length, CRC, autostart, and whether it ran at the last start
`stack`, `words`	the data stack at the last prompt; the dictionary
`board`	LEDs and button, both pin ports with the pins driven now, PWM channels, the 9 ADC codes, the temperature code, the accelerometer, the DDS (control and tuning) and the pin functions of the serial interfaces (`fn`: UART0 TX, RX, UART1 TX, RX, SPI0 SCK, MOSI, MISO, SPI1 SCK, MOSI, MISO, I2C SCL, SDA; bit 15 = on, bits 5..0 = the pin)
`source`, `store`, `erase`, `autostart on|off`	the saved source
`echo`, `restart [skip]`	a test echo; a restart

The details are in design.md. There is no HASE mode on the MAX1000.

5. Word reference

The kernel has 376 words. About a third of them are internal: they serve the interpreter, the compiler, the saved source and the app's requests (sections 5.24 and 5.25). Where a word differs from standard Forth, its entry says so.

5.1 Data stack

Word	Stack	Description
`dup`	`( x -- x x )`	Copy the top item.
`drop`	`( x -- )`	Remove the top item.
`swap`	`( x1 x2 -- x2 x1 )`	Exchange the top two items.
`over`	`( x1 x2 -- x1 x2 x1 )`	Copy the second item to the top.
`nip`	`( x1 x2 -- x2 )`	Remove the second item.
`tuck`	`( a b -- b a b )`	Copy the top item below the second.
`rot`	`( a b c -- b c a )`	Rotate the third item to the top.
`-rot`	`( a b c -- c a b )`	Rotate the top item to third place.
`2dup`	`( a b -- a b a b )`	Copy the top two items.
`2drop`	`( a b -- )`	Remove the top two items.
`?dup`	`( x -- x x | 0 )`	Copy the top item only if it is not zero.
`depth`	`( -- n )`	Number of items on the data stack, before `depth` itself pushes.
`.s`	`( -- )`	Show the stack without changing it: `<3> 1 2 3`, with the depth and the top on the right.

```
1 2 3 rot .s <3> 2 3 1  ok
```

5.2 Return stack

The return stack holds return addresses and, inside `do` loops, the loop index and limit. You can use it for temporary storage within one definition, but every `>r` needs a matching `r>` before the word ends or leaves a loop.

Word	Stack	Description
`>r`	`( x -- ) R: ( -- x )`	Move the top item to the return stack.
`r>`	`( -- x ) R: ( x -- )`	Move the top of the return stack back.
`r@`	`( -- x ) R: ( x -- x )`	Copy the top of the return stack.

These three are inline. They work inside definitions; typed directly at the terminal they would corrupt the interpreter's own return address.

5.3 Arithmetic

All arithmetic is on 16 bits and wraps around without warning.

Word	Stack	Description
`+`	`( n1 n2 -- n3 )`	Sum.
`-`	`( n1 n2 -- n3 )`	Difference n1 − n2.
`*`	`( n1 n2 -- n3 )`	Product, low 16 bits; one instruction (hardware multiplier).
`/`	`( n d -- quot )`	Quotient, rounded toward zero: `-7 2 /` gives −3.
`mod`	`( n d -- rem )`	Remainder, with the sign of n: `-7 2 mod` gives −1.
`/mod`	`( n d -- rem quot )`	Remainder and quotient (symmetric division).
`u/mod`	`( u d -- rem quot )`	Unsigned division: `-7 2 u/mod` gives 1 32764.
`*/`	`( n1 n2 n3 -- n4 )`	n1 × n2 / n3 with a 32-bit intermediate product, so the product may exceed 16 bits: `3000 1000 4096 */` gives 732. Rounds toward zero.
`um*`	`( u1 u2 -- ud )`	Full 32-bit unsigned product as two cells, the high cell on top: `-1 -1 um*` gives 1 −2 (65534).
`um/mod`	`( ud u -- rem quot )`	Divide a 32-bit unsigned number by u. The quotient must fit in 16 bits (high cell < u).
`1+`	`( n -- n+1 )`	Add one.
`1-`	`( n -- n-1 )`	Subtract one.
`2*`	`( n -- n*2 )`	Double.
`2/`	`( n -- n/2 )`	Arithmetic shift right; keeps the sign (`-8 2/` gives −4).
`negate`	`( n -- -n )`	Change the sign.
`abs`	`( n -- u )`	Absolute value. `-32768 abs` stays −32768 (as u: 32768).
`min`	`( a b -- min )`	Smaller of two signed numbers.
`max`	`( a b -- max )`	Larger of two signed numbers.
`umin`	`( u1 u2 -- u )`	Smaller of two unsigned numbers: `-1 3 umin` gives 3.

Division is done in software by shift-and-subtract (16 steps). Division by zero gives no error: `u/mod` then returns −1 as the quotient. `*/` and `um/mod` give a meaningless result if the quotient doesn't fit in 16 bits.

`*/` is the usual way to scale with fractions: `x 3300 4096 */` turns an ADC code into millivolts without losing the high bits of the product (that is what `mv@` does). `um*` gets its high cell from a dedicated CPU instruction (the high half of the hardware multiplier).

5.4 Logic and shifts

Word	Stack	Description
`and`	`( x1 x2 -- x3 )`	Bitwise AND.
`or`	`( x1 x2 -- x3 )`	Bitwise OR.
`xor`	`( x1 x2 -- x3 )`	Bitwise exclusive OR.
`invert`	`( x -- ~x )`	Invert all bits. Also turns a true flag into false and back.
`lshift`	`( x u -- x' )`	Shift left by u bits; u ≥ 16 gives 0.
`rshift`	`( x u -- x' )`	Logical shift right by u bits (fills with 0); u ≥ 16 gives 0.

5.5 Comparison

The comparison words return −1 (true) or 0 (false). `if`, `until` and `while` treat any non-zero value as true.

Word	Stack	Description
`=`	`( x1 x2 -- f )`	Equal.
`<>`	`( a b -- f )`	Not equal.
`<`	`( n1 n2 -- f )`	n1 less than n2, signed.
`>`	`( a b -- f )`	a greater than b, signed.
`u<`	`( u1 u2 -- f )`	Less than, unsigned: `-1 1 u<` is false.
`u>`	`( a b -- f )`	Greater than, unsigned.
`0=`	`( x -- f )`	x is zero; also a logical NOT.
`0<>`	`( x -- f )`	x is not zero.
`0<`	`( n -- f )`	n is negative.
`0>`	`( n -- f )`	n is positive.
`within`	`( n lo hi -- f )`	lo ≤ n < hi; works for signed and unsigned ranges.

5.6 Memory

Word	Stack	Description
`@`	`( a -- x )`	Fetch the cell at a (bit 0 of a ignored).
`!`	`( x a -- )`	Store x at a.

Word	Stack	Description
`c@`	`( a -- c )`	Fetch the byte at a.
`c!`	`( c a -- )`	Store the low byte of c at a.
`+!`	`( n addr -- )`	Add n to the cell at addr.
`?`	`( a -- )`	Print the cell at a, like `@ .`.
`cells`	`( n -- n*2 )`	Size of n cells in bytes.
`cell+`	`( a -- a+2 )`	Address of the next cell.
`aligned`	`( a -- a' )`	Round a up to an even address.
`cmove`	`( src dst n -- )`	Copy n bytes from src to dst, lowest address first.

```
variable x  5 x !  3 x +!  x ? 8  ok
```

5.7 Dictionary space and data RAM

New definitions and data are added at `here`, the next free byte of the code space (`$0000`..`$3FFF`). Larger data, such as sample buffers, belong in the **data RAM** (`$4000`..`$5BFF`), which `buffer:` hands out; it does not use up the space for code.

Word	Stack	Description
`here`	`( -- a )`	Address of the next free byte.
`allot`	`( n -- )`	Reserve n bytes at `here` (a negative n gives memory back).
`align`	`( -- )`	Make `here` even.
`,`	`( x -- )`	Align, store x at `here`, advance by one cell.
`c,`	`( c -- )`	Store a byte at `here`, advance by one byte.
`unused`	`( -- n )`	Free code space: `$4000` minus `here`.
`buffer: name`	`( n "name" -- )`	Reserve n bytes of data RAM (rounded up to even, so cells stay aligned); `name` then pushes their address. `data RAM full` if fewer than n bytes are left.
`unused-data`	`( -- n )`	Free data RAM in bytes.

A fresh kernel leaves 3,738 bytes of code space and 7,168 bytes of data RAM. `buffer:` rounds odd sizes up, so every buffer starts at an even address. The data RAM is not cleared, and a restart frees all of it, together with the words that named the buffers.

```
create table 1 , 2 , 4 , 8 ,
table 2 cells + ? 4  ok
100 buffer: buf  ok
buf . unused-data . 16384 7068  ok
```

5.8 Terminal input and output

Word	Stack	Description
`key`	`( -- c )`	Wait for a character from the terminal. Answers the app's requests while it waits.
`key?`	`( -- f )`	True if a character is waiting (does not take it). Answers the app's requests waiting before it.
`emit`	`( c -- )`	Send a character (waits while the 1 KB send FIFO is full).
`cr`	`( -- )`	New line (sends CR and LF).
`space`	`( -- )`	Send one space.
`spaces`	`( n -- )`	Send n spaces (nothing for n ≤ 0).
`type`	`( a n -- )`	Send n characters starting at a.
`.`	`( n -- )`	Print a signed number in the current base, then a space.
`u.`	`( u -- )`	Print an unsigned number, then a space: `-1 u.` gives 65535.
`." text"`	`( -- )`	Compile only: print `text` when the definition runs.
`.( text)`	`( -- )`	Print `text` immediately, in a definition or at the terminal.

Digits above 9 print as capital letters: `255 hex . decimal` gives `FF`.

```
: hello ." Hello, MAX1000!" cr ;
hello Hello, MAX1000!
 ok
```

5.9 Number base

Word	Stack	Description
`base`	`( -- a )`	Variable holding the base for number input and output (2..36).
`hex`	`( -- )`	Set the base to 16.
`decimal`	`( -- )`	Set the base to 10.

`5 2 base ! . decimal` prints 5 in binary: `101`. The base applies to input as well, so the 5 has to be typed before switching. After `2 base !`, typing `5` gives `5 ?`. An error doesn't reset `base`; type `decimal` to get back.

5.10 Board I/O

Word	Stack	Description
`led!`	`( n -- )`	Set LED1..LED8 from bits 0..7 of n (1 = on; bit 0 = LED1).
`led@`	`( -- n )`	Read back the LED state.
`led-on`	`( n -- )`	Switch LED n (1..8) on; the other LEDs stay as they are.
`led-off`	`( n -- )`	Switch LED n (1..8) off; the other LEDs stay as they are.

Word	Stack	Description
`led-toggle`	`( n -- )`	Change LED n (1..8): on becomes off, off becomes on.
`led-mask`	`( n -- mask )`	The bit of LED n in `led!` (1 for LED1 … 128 for LED8); `bad led` for n outside 1..8.
`btn@`	`( -- f )`	The USER button: true while pressed. Debounced (sampled every 10 ms).
`ms`	`( n -- )`	Wait n milliseconds (0..32767). Answers the app's requests while it waits.
`ms@`	`( -- n )`	Millisecond counter; wraps every 65.536 s.

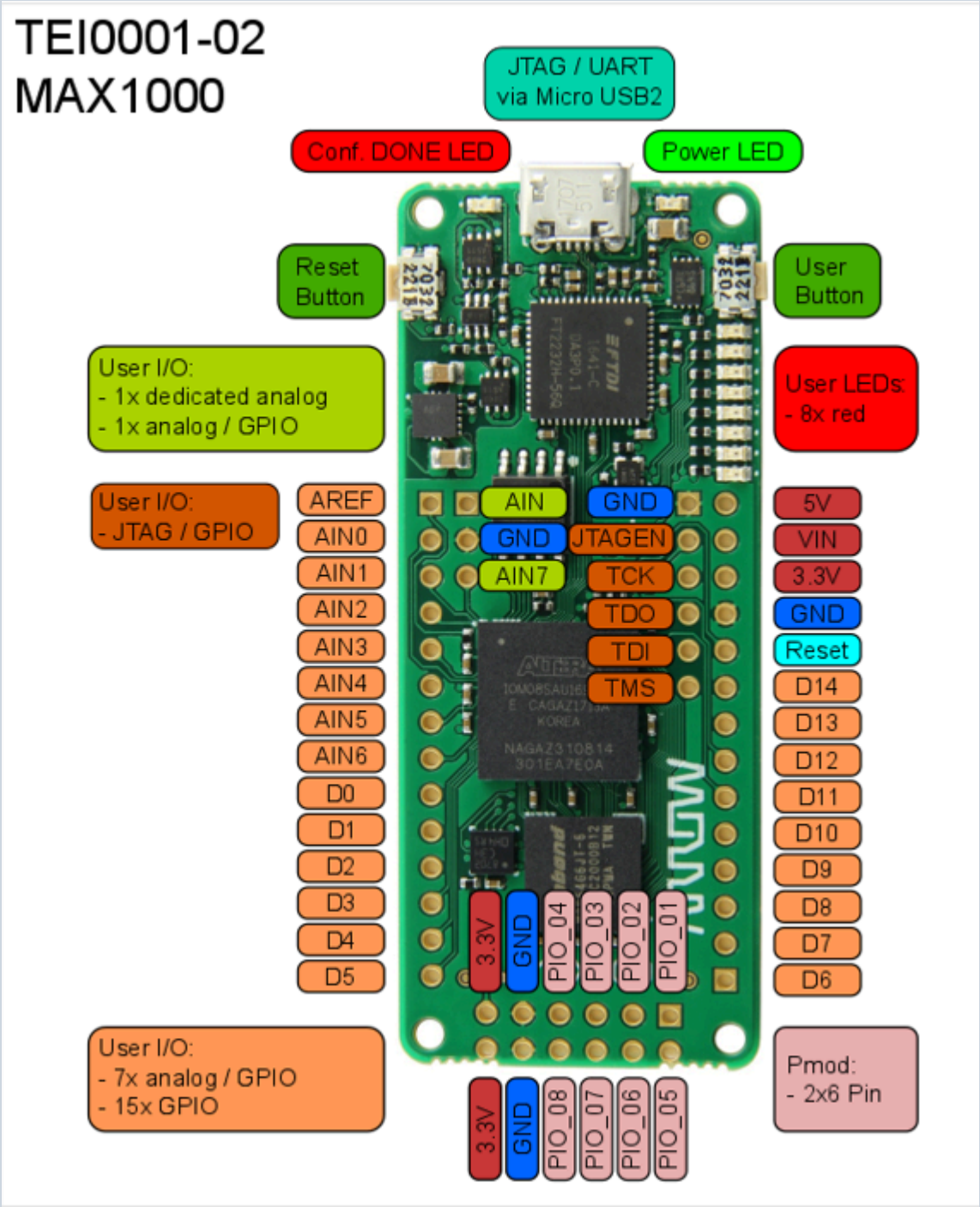
`led!` sets all eight LEDs at once; `led-on`, `led-off` and `led-toggle` change one LED and leave the others alone (they read the state with `led@`, change one bit and write it back). The LED numbers are those printed on the board, LED1..LED8. `ms` waits at least n ms and less than n + 1. `btn@` reads "not pressed" for the first 10 ms after a restart. The RESET button cannot be read (section 1).

```
$81 led!          \ LED1 and LED8 on, all others off
3 led-on          \ LED3 on as well: LED1, LED3 and LED8 are lit
1 led-off         \ LED1 off: LED3 and LED8 stay on
led@ .            \ prints 132 ($84)
: blink3 10 0 do 3 led-toggle 200 ms loop ;
btn@ .            \ hold the USER button while pressing Enter: prints -1
```

5.11 Digital pins

The digital pins are numbered after their labels on the board:

Pin numbers	Label	Header
0..14	D0..D14	the Arduino-style digital header
16..23	PIO_01..PIO_08	the Pmod header (PIO1..PIO8 in the app)



The MAX1000's headers (pinout image: Trenz Electronic, TEI0001-02). D0..D14 are Forth's pins 0..14, PIO_01..PIO_08 are pins 16..23; AIN, AIN0..AIN7 are the analog inputs 8 and 0..7 of `adc@` (section 5.12). Any of the 23 digital pins can carry a PWM channel, the DDS or a signal of a serial interface (sections 5.13 to 5.15). All I/O is 3.3 V only.

Pin 15 and pins 24 and up do not exist; the pin words answer `bad pin` for them. Every pin is an **input after start-up** and after Ctrl-R. `output` makes it drive the level of its output latch, which starts low.

Word	Stack	Description
`output`	`( n -- )`	Make pin n an output.
`input`	`( n -- )`	Make pin n an input again.
`high`	`( n -- )`	Set the pin's output latch to 1 (3.3 V when it is an output).

Word	Stack	Description
`low`	`( n -- )`	Set the output latch to 0.
`toggle`	`( n -- )`	Invert the output latch.
`pin!`	`( f n -- )`	`high` if f is true, otherwise `low`.
`pin@`	`( n -- f )`	Read the level at the pin: true = high. Also works on outputs, where it reads back the driven level.
`pin`	`( n -- mask a )`	Bit mask and port address of pin n; `bad pin` for numbers that are not pins.
`bset`	`( mask a -- )`	Set the mask bits in the cell at a (a general helper).
`bclr`	`( mask a -- )`	Clear the mask bits in the cell at a.
`owner`	`( n -- id )`	Who has pin n (section 5.15): 0 = nobody (the pin follows its latch and direction), 1..4 = PWM channel 0..3, 5, 6 = UART 0, 1, 7, 8 = SPI 0, 1, 9 = I2C, 10 = DDS.

`high` and `low` work on inputs too: they only set the latch. Setting the level first and then calling `output` makes the pin start at that level without a short glitch.

```
3 output  3 high            \ D3 to 3.3 V
3 pin@ . -1  ok
3 low  3 input              \ back to an input
```

No pull-ups. The pins have no pull-up or pull-down resistors, so **an open input reads at random**. A push button from a pin to GND needs a resistor of about 10 kΩ from the pin to 3.3 V. Alternatively, the FPGA can be built with its weak pull-ups switched on for all D and PIO pins (`scripts\build.ps1 -PullUps`, chapter 8); then an open input reads high and a button to GND needs no resistor. The pull-ups are fixed in the configuration; Forth cannot switch them.

Electrical. The pins are **3.3 V only; they are not 5 V tolerant.** Keep the current to a few mA, which is enough for an LED with a resistor of 330 Ω to 1 kΩ.

Speed. `toggle`, `high`, `low` and `pin@` take about 1 µs each, so a loop of `toggle` makes a square wave of a few hundred kHz. Each input is synchronized to the 48 MHz clock through two flip-flops.

Whole ports. The pins are bits of two 16-bit ports. Pin n is bit n mod 16 of port n / 16. Each port has three registers: pin levels at `$8010 + 8p`, the output latch at `$8012 + 8p`, and the direction at `$8014 + 8p` (1 = output); a fourth, `$8016 + 8p`, reads which pins are driven now, by their latch or by a peripheral. Writing whole ports changes several pins at once:

```
255 $8014 !                 \ D0..D7 as outputs
170 $8012 !                 \ D1, D3, D5, D7 high, the others low
$8010 @ 255 and .           \ read D0..D7 at once
```

5.12 Analog inputs and temperature

The MAX 10 contains a 12-bit **ADC**. It converts the nine analog inputs and the chip's temperature sensing diode continuously in the background, so reading a value takes no waiting; each input is refreshed many times per

millisecond. The reference is the 3.3 V supply: 0..3.3 V give the codes 0..4095.

The inputs are numbered after their labels on the board:

`adc@` input	0	1	2	3	4	5	6	7	8
Board label	AIN0	AIN1	AIN2	AIN3	AIN4	AIN5	AIN6	AIN7	AIN
ADC channel	8	2	5	1	3	4	7	6	0

AIN is the MAX 10's dedicated analog input; AIN0..AIN7 are dual-purpose pins. The table `ain` maps the inputs to the converter's channels, which `adc-ch@` reads directly.

Word	Stack	Description
`adc@`	`( n -- u )`	Raw result of input n (0..8, the table above): 0..4095 for 0 to 3.3 V. `bad input` for other numbers.
`mv@`	`( n -- mv )`	The input's voltage in millivolts: 0..3299.
`adc-ch@`	`( c -- u )`	Raw result of ADC channel c, 0..8, in the converter's own numbering; 9 is the temperature diode.
`ain`	`( -- a )`	Table of 9 bytes: the ADC channel of each input (used by `adc@`).
`temp-code@`	`( -- u )`	Raw code of the temperature sensing diode, a running average of 64 samples. It falls as the chip warms.
`temp@`	`( -- n )`	Chip temperature in tenths of a degree Celsius: 346 means 34.6 °C.
`.temp`	`( -- )`	Print the chip temperature, for example `34.6 C`.

```
3 mv@ . 1650  ok           \ for example: a potentiometer half way
.temp 34.6 C  ok
temp-code@ . 3640  ok
```

The readings shown come from the simulator (chapter 8); on the board your values differ.

Range. Keep every analog input between 0 and 3.3 V. Voltages near 3.3 V read as 3299 mV.

Open inputs. An input with nothing connected does not read 0: it shows the charge left on the converter by the channel converted before it. Only driven inputs give meaningful values. To make high-impedance sources (a potentiometer of tens of kΩ) read right, the converter samples each channel four times in a row and keeps only the last result.

Temperature. `temp@` interpolates linearly between two points of Intel's table for the diode, −40 °C = 3798 and 96 °C = 3510, so it is approximate: expect a few degrees of error. It is the die temperature, typically somewhat above the room.

5.13 PWM

Four PWM channels, 0..3, make square waves of variable frequency and duty cycle in the FPGA, independently of Forth: once set, a channel runs on its own, also while Forth computes or waits for input. Each channel drives **any digital pin** (section 5.11).

Word	Stack	Description
`pwm-pin`	`( pin ch -- )`	Route channel ch to the pin and start it. The pin becomes an output driven by the channel; `bad pin` for numbers that are not pins, `pin taken` if another channel, the DDS or a serial interface has it. A channel that had another pin lets it go.
`pwm-off`	`( ch -- )`	Stop the channel. Its pin returns to its own latch and direction (an input after start-up).
`pwm-hz`	`( u ch -- )`	Frequency in Hz, 1..65535; `bad frequency` for 0.
`pwm-khz`	`( u ch -- )`	Frequency in kHz, 1..24000; `bad frequency` outside.
`pwm-duty`	`( permille ch -- )`	Duty cycle in ‰ (tenths of a percent), 0..1000: 0 is always low, 1000 always high; values outside are clamped.
`pwm-clocks`	`( ud ch -- )`	The period directly, in 48 MHz clocks (20.8 ns), as a double number: low cell, then high cell. At least 2.
`pwm`	`( ch -- a )`	Register address of channel ch; `bad channel` for ch above 3.
`ud/u`	`( ud u -- ud' )`	Unsigned double divided by an unsigned single, giving a double quotient (used by `pwm-hz`).

The order does not matter: frequency, duty and pin can be set in any order and changed at any time. **A change takes effect at the end of the running cycle,** so a pulse is never cut short and the frequency or duty can be swept while the channel runs. After start-up, and after Ctrl-R, every channel is off with a period of 2 clocks (24 MHz) and duty 0.

```
1000 0 pwm-hz  250 0 pwm-duty  3 0 pwm-pin    \ D3: 1 kHz, 25 % high
500 0 pwm-duty                                \ now 50 %
2 1 pwm-hz  100 1 pwm-duty  4 1 pwm-pin       \ D4: 2 Hz, 50 ms pulses
0 pwm-off  1 pwm-off
```

Resolution. The period is counted in clocks of 20.8 ns, so a frequency is exact when 48,000,000 divided by it is a whole number; otherwise the period is rounded down (`pwm-khz` divides 48,000 by the kHz value). The duty cycle is set in steps of 1/65536 of the period, and the pulse is a whole number of clocks: at 1 kHz (48,000 clocks) the duty can be set to about 0.002 %, at 1 MHz (48 clocks) only in steps of about 2 %. For a fine duty cycle, keep the frequency well below 1 MHz.

A pin belongs to one channel at a time (section 5.15). The pin's level can be read with `pin@` like any output.

Checking it without an instrument. A PWM pin filtered by a resistor and a capacitor (10 kΩ from the pin to an analog input, 1 µF from that input to GND) gives a DC voltage of duty × 3.3 V, which the ADC reads (section 6.4).

Registers. Channel c is at `$8080` + 8c:

Address	Register
+0	period, low 16 bits (taken when the high half is written)
+2	period, high 16 bits; writing it sets the period in clocks
+4	duty, in 1/65536 of the period; `$FFFF` = always high
+6	bit 15 = on, bits 5..0 = pin number

```
1000 0 pwm-hz  0 pwm @ u. 48000  ok
```

5.14 DDS square-wave generator

For a square wave of an exact, finely set frequency there is a DDS (direct digital synthesis) generator: a 32-bit phase accumulator on its own 192 MHz clock, whose top bit is the output. It drives **any digital pin**, like a PWM channel, and runs on its own once set.

Word	Stack	Description
`dds-hz`	`( u -- )`	Frequency in Hz, 0..65535.
`dds-khz`	`( u -- )`	Frequency in kHz, 0..65535 (up to 65.5 MHz).
`dds-tuning`	`( ud -- )`	The phase step directly: f = ud × 192 MHz / 2^{32} (low cell, then high cell).
`dds-pin`	`( pin -- )`	Route the DDS to the pin and start it at phase 0; `bad pin` for numbers that are not pins, `pin taken` if a PWM channel or a serial interface has it.
`dds-off`	`( -- )`	Stop it; the pin returns to its own latch and direction.
`dds-sync`	`( -- )`	Wait until the generator has taken over the last change (the others do this themselves).

The frequency resolution is 192 MHz / 2^{32} = 0.0447 Hz, so `1 dds-hz` gives 0.98 Hz and `1000 dds-hz` 1000.01 Hz: the error is below 0.05 Hz everywhere, which matters only at very low frequencies. The edges fall on a 5.2 ns grid. A frequency that does not divide 192 MHz evenly is therefore made of periods that differ by one 5.2 ns step, which averages to the exact value (at 1 MHz: ±0.5 % jitter per period, at 10 MHz ±5 %). Above about 20 MHz the shape is limited by the pin and its wiring. A change of frequency takes effect at once and keeps the phase running, so a sweep has no gaps; `dds-pin` restarts the phase.

```
1000 dds-hz  3 dds-pin        \ D3: 1 kHz
1500 dds-khz                  \ now 1.5 MHz
dds-off
```

`dds-hz` and `dds-khz` compute the phase step hz × 2^{32} / 192,000,000 = hz × 2^{20} / 46875 by long division, 16 bits at a time with `um/mod`, and round it (`(dds-hz) ( ud -- )` takes the frequency in Hz as a double number; `dds-d` is the constant 46875). The DDS registers are at `$8060`:

Address	Register
`$8060`	tuning, low 16 bits (taken when the high half is written)
`$8062`	tuning, high 16 bits; writing it sets the frequency
`$8064`	bit 15 = on, bits 5..0 = pin number; read: bit 14 = a change not yet taken over

```
1000 dds-hz  $8060 @ hex . decimal 5762  ok
```

5.15 Serial interfaces: UART, SPI, I2C

Two **UARTs**, two **SPI** masters and an **I2C** master run in the FPGA and put each of their signals on **any digital pin** (section 5.11). They work on their own once started: a UART receives into its buffer while Forth computes, and an SPI or I2C transfer runs while Forth waits for it.

Pins belong to one user. A pin given to an interface, a PWM channel or the DDS belongs to it until that one is closed or switched off: its latch and direction no longer count, and nobody else can take it (`pin taken`). The open words check all their pins first, so a refused open changes nothing. `owner` tells who has a pin. A restart, Ctrl-R or `cold` closes everything.

Word	Stack	Description
`owner`	`( pin -- id )`	0 = nobody, 1..4 = PWM channel 0..3, 5, 6 = UART 0, 1, 7, 8 = SPI 0, 1, 9 = I2C, 10 = DDS. `bad pin` for numbers that are not pins.

UART u = 0 or 1: 8 data bits, no parity, 1 stop bit (8N1). Each has a 1 KB receive buffer and a 1 KB send buffer in the FPGA, so nothing is lost while Forth does something else for a while.

Word	Stack	Description
`uart-open`	`( ud-baud tx rx u -- )`	Open UART u at the baud rate, a double number (`115200.`), with TX and RX on the given pins. `bad baud` for 0 or less than 3 Bd; above 6,000,000 Bd it runs at 6 MBd. Opening an open UART again moves it to the new pins and rate.
`uart-close`	`( u -- )`	Close it and empty its buffers; its pins return to their own latch and direction.
`uart-emit`	`( c u -- )`	Send a byte; waits only while the send buffer is full.
`uart-type`	`( a n u -- )`	Send n bytes from a.
`uart-key?`	`( u -- f )`	True if a byte has arrived.
`uart-key`	`( u -- c )`	The next byte, waiting for it. The app's requests are answered while it waits.
`uart-status`	`( u -- bits )`	Bit 2 = everything sent, bit 3 = bytes were lost (the receive buffer was full), bit 4 = a framing error (wrong baud rate, or a break). Reading clears bits 3 and 4.

The UART counts **48,000,000 / baud** clocks per bit, rounded: 115,200 Bd gives 417 clocks (115,108 Bd, 0.08 % slow), 9600 gives 5000 (exact), 230,400 gives 208 and 921,600 gives 52 (both 0.16 % fast), 1,000,000 and 3,000,000 are exact. A UART tolerates about 2 %, so every usual rate works. The levels are 3.3 V: a PC's RS-232 port needs a level converter, a USB serial adapter for 3.3 V can be connected directly (its TX to our RX, its RX to our TX, GND to GND).

```
115200. 0 1 0 uart-open      \ UART 0: TX on D0, RX on D1
72 0 uart-emit 105 0 uart-emit
0 uart-key? .                \ -1 if something came back
0 uart-close
```

A **loopback** needs no wire: TX and RX on the same pin receive what is sent, since a pin reads back the level it drives (`115200. 7 7 0 uart-open`).

SPI s = 0 or 1: master, modes 0..3, 8 or 16 bits per transfer, most significant bit first unless `spi-lsb` says otherwise. After start-up: 1 MHz, mode 0, 8 bits, MSB first; these settings are kept when the interface is closed and opened again. **Chip selects are ordinary pins**: make the pin an output and drive it low around the

transfers. (The words `spi` and `spi-cs` of section 5.17 belong to the on-board flash and accelerometer, not to these.)

Word	Stack	Description
`spi-open`	`( sck mosi miso s -- )`	Open SPI s with SCK, MOSI and MISO on the given pins.
`spi-close`	`( s -- )`	Close it; its pins return to their own latch and direction.
`spi-khz`	`( khz s -- )`	Clock in kHz, 1..12000; `bad frequency` outside. The clock is 24,000 / n kHz for a whole n, the nearest one not above the value: 12000, 8000, 6000, 4800, 4000, …, 1000 (exact), …
`spi-mode`	`( mode s -- )`	Mode 0..3: bit 1 = CPOL (the clock's idle level), bit 0 = CPHA (0: data taken on the first edge, 1: on the second).
`spi-lsb`	`( f s -- )`	True: least significant bit first.
`spi-bits`	`( n s -- )`	8 or 16 bits per transfer; `bad bits` for anything else.
`spi-xfer`	`( out s -- in )`	Send out and return what came back at the same time (8 or 16 bits).
`spi-write`	`( a n s -- )`	Send n bytes from a; what comes back is dropped.
`spi-read`	`( a n s -- )`	Receive n bytes into a, sending zeros.

```
10 output 10 high               \ D10 = chip select, idle high
13 11 12 0 spi-open             \ SCK D13, MOSI D11, MISO D12
1000 0 spi-khz
10 low $9F 0 spi-xfer drop 0 0 spi-xfer . 10 high
```

A transfer at 1 MHz takes 8 µs per byte, at 12 MHz under 1 µs. MISO is sampled at the end of each bit, as late as possible, which also suits slow devices and long wires.

I2C: master, 7-bit addresses, 1..1000 kHz, clock stretching by the device (up to 22 ms, then `i2c timeout`). SCL and SDA are **open drain**: the FPGA pulls them low or lets go, so both lines need **pull-up resistors** to 3.3 V. Most I2C modules have them; otherwise use 4.7 kΩ from each line to 3.3 V. (The weak pull-ups of the `-PullUps` build, section 5.11, are too weak except for slow, short connections.)

Word	Stack	Description
`i2c-open`	`( khz scl sda -- )`	Open with SCL and SDA on the given pins; 100 = standard, 400 = fast mode; the clock is at most the value given. `bad frequency` outside 1..1000.
`i2c-close`	`( -- )`	Close it; the pins are free again.
`i2c!`	`( b reg addr -- )`	Write byte b into register reg of the device at addr.
`i2c@`	`( reg addr -- b )`	Read register reg of the device at addr (a write of reg, a repeated start, a read).
`i2c-probe`	`( addr -- f )`	True if a device acknowledges addr.
`i2c-scan`	`( -- )`	Print the addresses $08..$77 that answer, in hex.
`i2c-start`	`( -- )`	A start condition (or a repeated start).

Word	Stack	Description
`i2c-stop`	`( -- )`	A stop condition.
`i2c-addr`	`( addr rw -- ack? )`	A start, then the address byte: rw 0 = write, 1 = read. True if the device acknowledged.
`i2c-write`	`( b -- ack? )`	Send a byte; true if the device acknowledged.
`i2c-read`	`( ack? -- b )`	Receive a byte; true: acknowledge it (more will be read), false: it is the last.

`i2c!` and `i2c@` stop with `no ack` when the device does not acknowledge (wrong address, nothing connected, no pull-ups); they send a stop first. Devices with other register layouts are reached with the low-level words, for example two bytes from register 0 of a sensor at `$48`:

```
: rd16 ( -- n )
  $48 0 i2c-addr drop 0 i2c-write drop
  $48 1 i2c-addr drop -1 i2c-read 8 lshift 0 i2c-read or i2c-stop ;
```

Registers. UART u at `$80A0` + 16u, SPI s at `$80C0` + 16s, I2C at `$80E0`:

Offset	UART	SPI	I2C
+0	read: bit 8 = a byte, bits 7..0 the byte (the read takes it); write: send	write: start a transfer; read: the value received last	write: bits 10..8 command (1 start, 2 stop, 3 write byte, 4 read + ACK, 5 read + NACK), bits 7..0 byte; read: bit 15 busy, bit 9 timeout, bit 8 ACK, bits 7..0 byte
+2	status (as `uart-status`, bit 0 = a byte, bit 1 = room to send); write: clear bits 3, 4	bit 0 = busy	clocks per quarter SCL period
+4	clocks per bit, bits 15..0 (at least 8)	bit 0 CPHA, bit 1 CPOL, bit 2 LSB first, bit 4 16 bits	bit 15 on, bits 13..8 SDA pin, bits 5..0 SCL pin
+6	clocks per bit, bits 23..16	clocks per half SCK period (at least 2)	–
+8	bit 15 on, bits 13..8 RX pin, bits 5..0 TX pin	bit 15 on, bits 13..8 MOSI pin, bits 5..0 SCK pin	–
+10	–	bits 5..0 MISO pin	–

The words keep the owner table (`pin-owner`, section 4.3) in step with these registers; writing the pin registers directly bypasses it.

5.16 Accelerometer

The MAX1000 carries an ST **LIS3DH** three-axis accelerometer on its own SPI pins, driven by the same SPI master as the flash. `cold` switches it on (`accel-on`): 100 readings per second on all three axes, ±2 g, high resolution, so one digit is 1 mg.

Word	Stack	Description
`accel@`	`( -- x y z )`	The acceleration on the three axes in mg (thousandths of g). Lying flat, components up, the board reads about 0 0 1000.
`accel?`	`( -- f )`	True if the accelerometer answers (its WHO_AM_I register reads `$33`).
`accel-on`	`( -- )`	Set it up as above: 100 Hz, X/Y/Z on, block data update, high resolution, ±2 g.
`acc@`	`( reg -- c )`	Read LIS3DH register reg.
`acc!`	`( c reg -- )`	Write c to LIS3DH register reg.

```
accel? . -1  ok
accel@ rot . swap . . 0 0 1000  ok          \ x y z, the board lying flat
```

A board at rest measures gravity: 1 g (about 1000 mg) shared between the axes as you tilt it. The values are updated 100 times a second; `accel@` returns the latest set. With `acc!` other settings of the LIS3DH are possible (see its data sheet); for example `$98 $23 acc!` selects ±4 g, after which `accel@` returns units of 2 mg. `accel-on` (or a restart) sets ±2 g again.

5.17 Serial flash

The MAX1000 has a Winbond **W25Q64** serial flash of 8 MB. The FPGA does not configure from it (the MAX 10 has its own internal flash), so all of it is available to Forth. The saved source uses two areas (section 4.7); the rest is free for your own data:

Flash address	Use
`$000000`–`$000FFF`	the saved source's header
`$001000`–`$00FFFF`	free
`$010000`–`$01FFFF`	the saved source's text
`$020000`–`$7FFFFF`	free (8,064 KB)

Flash addresses have 24 bits and are given as two cells, `( lo hi )`: lo = bits 15..0, hi = bits 23..16. `$123456` is `$3456 $12`; `0 $10` is `$100000`, 1 MB into the flash.

Word	Stack	Description
`f-read`	`( lo hi -- )`	Start reading at the address. Then each `0 spi` returns the next byte; the read runs on across the whole flash. End it with `f-end`.
`f-end`	`( -- )`	Deselect the flash (and the accelerometer): ends a read or a command.
`f-prog`	`( a n lo hi -- )`	Program n bytes from RAM address a into the flash. They must stay within one 256-byte page (lo's low byte + n ≤ 256), and the area must be erased. Waits until done.
`f-erase`	`( lo hi -- )`	Erase the 4 KB sector that holds the address (all bytes become `$FF`). Waits until done.
`f-erase64`	`( lo hi -- )`	Erase the 64 KB block that holds the address. Waits until done.

Word	Stack	Description
`flash-id`	`( -- mfr type size )`	The JEDEC ID: `$EF $40 $17` for the W25Q64.
`spi`	`( c -- c' )`	Exchange one byte with the selected device: send c, return the byte received.
`spi-cs`	`( f -- )`	Select the flash if bit 0 of f is set, else deselect it.
`f-cmd`	`( c -- )`	Select the flash and send the command byte c.
`f-adr`	`( lo hi -- )`	Send a 24-bit address, high byte first.
`f-wren`	`( -- )`	Send *write enable* (needed before every program or erase command).
`f-wait`	`( -- )`	Wait until the flash has finished programming or erasing.
`crc-reset`	`( -- )`	Start a new CRC-32.
`crc+`	`( c -- )`	Add a byte to the CRC.
`crc@`	`( -- lo hi )`	The CRC-32 so far, as two cells. It is the standard CRC-32 (as zlib computes it).

Programming can only turn 1 bits into 0 bits: **erase before you program.** Programming over old data without erasing gives the AND of the old and the new bytes. An erase typically takes tens of milliseconds (4 KB) to a few hundred (64 KB), programming a page about a millisecond; Forth answers no app requests meanwhile.

```
hex flash-id . . . decimal 17 40 EF  ok
variable v  4711 v !  ok
0 $10 f-erase  v 2 0 $10 f-prog  ok          \ 2 bytes to $100000
: f@ ( lo hi -- x ) f-read 0 spi 0 spi f-end 8 lshift or ;  ok
0 $10 f@ . 4711  ok
: crc-test crc-reset 58 49 do i crc+ loop crc@ ;  ok    \ the bytes "123456789"
hex crc-test u. u. decimal CBF4 3926  ok
```

Don't damage the saved source. `f-erase` or `f-erase64` at addresses below `$020000` can destroy the saved source or its header; `f-erase64` at `0 0` erases both the header and the free area behind it.

The SPI master is shared between the flash and the accelerometer: selecting one deselects the other. While a read started by `f-read` is open (until `f-end`), call no accelerometer word (`accel@`, `acc@`, …) and no other flash word, or the read ends. The app cannot get in the way: while the flash or the accelerometer is selected, the kernel answers no request (they wait until `f-end`), so you may print or wait inside an open read. `key?` answers false while such a request waits at the head of the input, and terminal input behind the request waits as well, so end the read before you read the keyboard.

5.18 Saved source

These words work on the source kept in the flash (section 4.7). The app does the same with its saved-source box (section 2.7).

Word	Stack	Description
`.source`	`( -- )`	List the saved source.
`load-source`	`( -- )`	Run the saved source now, after the rest of the current line.

Word	Stack	Description
`autostart`	`( f -- )`	Switch the autostart on (true) or off (false). `no saved source` if there is no valid source.
`erase-source`	`( -- )`	Delete the saved source (its header).
`source?`	`( -- len | 0 )`	The length of the saved source if its header and CRC are right, else 0.
`store`	`( len flags -- )`	Receive and save a source: prints `ready`, then reads len raw bytes from the serial port and stores them with the flags (bit 0 = autostart). Prints `stored n bytes`, `store timeout` (2 s without a byte) or `store failed`. Used by `mforth.py --store`.

`store` is not meant to be typed: the text must follow at once and byte for byte. From the command line:

```
python tools/mforth.py -p COMx --store examples/bench.fs                # save, autostart on
python tools/mforth.py -p COMx --store examples/bench.fs --no-autostart  # save only
python tools/mforth.py -p COMx --skip                                    # restart without it
```

```
source? . 711  ok                \ for example
0 autostart  ok
.source \ Bench demo for the wired test environment: potentiometer on AIN3, button on
...
```

5.19 Defining words

Word	Stack	Description
`: name`	`( -- )`	Start a new definition called `name` and switch to compiling.
`;`	`( -- )`	End the definition, make it findable, switch back to interpreting.
`variable name`	`( -- )`	Create a variable (one cell, initially 0). `name` then pushes its address.
`constant name`	`( n -- )`	Create a constant. `name` then pushes n.
`create name`	`( -- )`	Create a word that pushes the address of the data space following it; reserve that space with `allot`, `,` or `c,`.
`buffer: name`	`( n "name" -- )`	Create a word that pushes the address of n bytes of data RAM (section 5.7).
`immediate`	`( -- )`	Make the newest word immediate: it runs during compilation instead of being compiled.

```
variable count
10 constant ten
create buffer 20 allot
: bump 1 count +! ;
```

The name of a definition can't be found until `;`. So a word can be redefined in terms of its old version:

```
: sq dup * ;
: sq sq 1+ ;      \ the inner sq is the old one
5 sq . 26  ok
```

Use `recurse` to call the word being defined.

5.20 Control structures

These are immediate and work **only inside colon definitions**.

Structure	Description
`f if ... then`	Run the code if f is true.
`f if ... else ... then`	Choose between two branches.
`begin ... f until`	Repeat until f is true (runs at least once).
`begin ... f while ... repeat`	Repeat while f is true, test in the middle.
`begin ... again`	Repeat forever (leave with `exit`, or Ctrl-R from the terminal).
`limit start do ... loop`	Counted loop: the index runs from start to limit − 1.
`limit start do ... n +loop`	Like `loop`, but adds n to the index; ends when the index is ≥ limit. Counts upward only.
`exit`	Return from the current word immediately.
`recurse`	Call the word being defined.

Words for use inside `do` loops:

Word	Stack	Description
`i`	`( -- n )`	Index of the innermost loop.
`j`	`( -- n )`	Index of the next outer loop (in the same definition).
`unloop`	`( -- )`	Remove the loop parameters from the return stack. Required before `exit` inside a loop.

`do` always runs the body at least once, and the index runs up to the limit with wrap-around. So `0 0 do ... loop` runs 65536 times. There is no `?do` and no `leave`.

```
: fact ( n -- n! ) dup 1 > if dup 1- recurse * then ;
7 fact . 5040  ok
: evens 10 0 do i . 2 +loop ;
evens 0 2 4 6 8  ok
: table 3 1 do 3 1 do j i * . loop loop ;
table 1 2 2 4  ok
: find3 10 0 do i 3 = if ." found" unloop exit then loop ." none" ;
```

5.21 Compiler and execution

Word	Stack	Description
`' name`	`( -- xt | 0 )`	Execution token of `name`; **0 if not found** (standard Forth signals an error).
`['] name`	`( -- )`	Compile only: compile `name`'s xt as a literal.
`execute`	`( xt -- )`	Run the word with this execution token. `0 execute` runs `cold` (section 4.6).

Word	Stack	Description
`[`	`( -- )`	Switch to interpreting inside a definition (immediate).
`]`	`( -- )`	Switch back to compiling.
`literal`	`( n -- )`	Compile n as a number (immediate). Use it with `[` and `]` to compute values at compile time.
`compile,`	`( xt -- )`	Compile a call to xt.
`state`	`( -- a )`	Variable: −1 while compiling, 0 while interpreting.

```
: five [ 2 3 + ] literal ;  ok
five . 5  ok
' five execute . 5  ok
: shout ." loud" ; immediate
: quiet shout ; loud ok        \ shout ran while quiet was compiled
```

5.22 Comments

Word	Description
`( text)`	Ignore everything up to the closing `)`. Commonly used for stack comments.
`\ text`	Ignore the rest of the line.

Both work at the terminal and inside definitions. Like every Forth word, each needs a space after it.

5.23 Interpreter and system

Word	Stack	Description
`words`	`( -- )`	List all words, newest first (internal words included).
`quit`	`( -- )`	Return to the interpreter loop (the data stack is kept).
`abort`	`( -- )`	Clear the data stack and return to the interpreter loop; stops a running saved source.
`cold`	`( -- )`	Restart in software: the kernel's dictionary, PWM, DDS and serial interfaces off, decimal base, banner, LED1 on, the saved source if autostart is on, interpreter loop. Pins keep their state (section 4.6).

5.24 Variables and constants used by the system

Word	Content
`dp`	Dictionary pointer: the value `here` returns.
`latest`	Header address of the newest word.
`dp0`, `latest0`	`dp` and `latest` of the fresh kernel; `cold` restores them.
`dp2`	Next free byte of the data RAM (`buffer:`).

Word	Content
`data0`, `data-end`	Constants: `$4000` and `$5C00`, the data RAM `buffer:` hands out.
`state`	Compile state (section 5.21).
`base`	Number base (section 5.9).
`tib`	Constant: the terminal input buffer, `$5D00` (82 bytes).
`#tib`	Number of characters in the current input line.
`>in`	Parse position in `tib`.
`'a`, `'n`	Address and length of the name being searched by `find-name`.
`sbuf`, `sn`	Buffer and count used by `.s`.
`ssnap`, `sdepth`	The copy of the data stack `snap` makes before each line (20 cells, bottom first) and its depth, −1 while copying; the `stack` request sends them.
`starts`	Starts since the configuration was loaded (the app's status line shows it).
`last-start`	0: no saved source ran at the last start, 1: it ran, 2: it was skipped.
`kcheck`	The sum of the kernel image's cells, set by the cross-compiler (part of the `version` answer).

5.25 Internal words

These words are visible in `words` and can be used, but they mainly exist for the kernel.

Loop and string runtime: these are compiled by `loop`, `+loop`, `."` and `.`.

Word	Stack	Description
`(loop)`	`( -- done? )`	Increment the loop index; true when it reached the limit.
`(+loop)`	`( n -- done? )`	Add n to the loop index; true when it is ≥ the limit.
`(.")`	`( -- )`	Print the counted string stored after the call and continue behind it.
`(s")`	`( -- a n )`	Push the counted string stored after the call (used by the kernel's own `s"` strings, which only the cross-compiler understands).
`(u.)`	`( u -- )`	Print u without a trailing space.
`digit`	`( u -- c )`	Convert 0..35 to the character `0`..`9`, `A`..`Z`.

Parsing the current input line:

Word	Stack	Description
`parse-name`	`( -- a n )`	Skip spaces, return the next space-delimited word (n = 0 at end of line).
`parse`	`( c -- a n )`	Return the text up to character c and skip past c.
`more?`	`( -- f )`	True if unparsed characters remain on the line.
`in-char`	`( -- c )`	The character at the parse position.

Word	Stack	Description
`/string`	`( a n k -- a+k n-k )`	Remove k characters from the front of a string.
`upper`	`( c -- c' )`	Convert a lower-case letter to upper case.

Dictionary search:

Word	Stack	Description
`find-name`	`( a n -- h | 0 )`	Find a visible word by name (case-insensitive), newest first.
`match?`	`( h -- f )`	True if header h is visible and matches the name in `'a/'n`.
`same?`	`( a1 a2 n -- f )`	Compare two strings of length n, case-insensitive.
`name>count`	`( h -- a n )`	Name string of a header.
`name>xt`	`( h -- xt )`	Execution token of a header.

Numbers:

Word	Stack	Description
`number?`	`( a n -- x f )`	Convert a string with optional `-` and `$` prefix; f is true on success.
`>number`	`( acc a n -- acc' a' n' )`	Accumulate digits into acc until a non-digit; returns the rest of the string.
`digit?`	`( c -- n f )`	Value of a digit character and whether it is valid in `base`.
`dnumber?`	`( a n -- lo hi f )`	Convert a string with a trailing dot (and an optional `$`) to an unsigned double number.
`>dnumber`	`( lo hi a n -- lo' hi' f )`	Accumulate the digits into a double number; f is true if all were digits.
`d*base+`	`( lo hi n -- lo' hi' )`	Multiply a double number by `base` and add n.
`m+u`	`( ud u -- ud' )`	Add a single to a double number.

Compiler:

Word	Stack	Description
`header`	`( a n -- )`	Lay down a header with this name and make it `latest`.
`flags!`	`( mask -- )`	OR mask into the count byte of `latest`.
`hide`	`( -- )`	Mark `latest` hidden (used by `:`).
`reveal`	`( -- )`	Clear the hidden flag of `latest` (used by `;`).
`compile-call`	`( xt -- )`	Compile a call instruction to xt.
`inline,`	`( xt -- )`	Copy xt's instructions up to its return into the definition.
`>resolve`	`( orig -- )`	Patch the jump at orig to point to `here` (used by `then`, `else`, `repeat`).

Interpreter:

Word	Stack	Description
`snap`	`( -- )`	Copy the data stack into `ssnap` and its depth into `sdepth`, leaving the stack as it was; `quit` runs it before each line, for the app's stack view (section 2.6).
`accept-line`	`( -- )`	Read one line into `tib`, with echo and backspace (no echo while the saved source is read).
`interpret`	`( -- )`	Interpret or compile the rest of the line.
`interpret-word`	`( a n -- )`	Handle one word: execute, compile, inline, or convert as a number.
`?stack`	`( -- )`	Check for stack underflow or overflow (section 3.4).

Serial port:

Word	Stack	Description
`rx?`	`( -- f )`	A byte is waiting in the receive FIFO (terminal byte or request).
`soh?`	`( -- f )`	The next byte is a SOH, the start of an app request.
`tx`	`( c -- )`	Send a byte as it is (no doubling of 0x01), answering requests while the send FIFO is full.

Serial interfaces and pin owners (section 5.15): `pin-owner` (the table, 32 bytes at `$5E20`), `(free) ( pin id -- pin )` (refuses a pin another owner has), `claim ( pin id -- )`, `(claim) ( pin id -- pin )`, `release ( id -- )` (frees every pin of that owner), `units-off ( -- )` (what `cold` runs), `cid` (the owner being opened), `uart-a ( u -- a )` and `spi-a ( s -- a )` (register addresses; `bad uart`, `bad spi`), `uart-div ( ud-baud u -- )` (sets the divider), `d2/u ( ud -- ud' )` (halves a double number), `spi-ctl ( bits mask s -- )` (changes control bits), `(i2c) ( cmd -- status )` (runs one I2C command and waits) and `(ack?) ( f -- )` (stops and reports `no ack`).

Accelerometer: `acc-sel ( -- )` and `acc-end ( -- )` select and deselect the LIS3DH; `mg ( lo hi -- mg )` turns its two output bytes into mg.

Saved source. `autoload ( -- )` is what `cold` runs: it checks the source (`scan-source ( -- )`, which reads the header with `hdr@ ( -- )`, checks it with `hdr-ok? ( -- f )` and the text with `text-crc ( len -- lo hi )`), waits up to 200 ms for an ESC (`esc? ( -- f )`), reads the button and starts the load (`(load) ( len -- )`). During the load, `key` takes its bytes from `fkey ( -- c )`, which refills `lbuf` 256 bytes at a time with `lfill ( -- )`, and `echo? ( -- f )` is false. `store` uses `receive ( len -- f )`, `page-in ( n -- f )`, `rx-t ( -- c | -1 )` (a byte within 2 s), `make-hdr ( flags -- )` and `write-hdr ( -- )`. Their variables and buffers: `hdr`, `pbuf`, `lbuf`, `fsrc`, `fcr`, `loff`, `lpos`, `lend`, `slen`, `soff`, `dl`, and the state found by the last check: `s-state` (0 none, 1 stored, 2 damaged), `s-len`, `s-auto`, `s-crclo`, `s-crchi`. `skip1` makes the next start skip the saved source (the `restart skip` request).

Service requests (section 4.8). `service? ( -- )` answers the requests at the head of the receive FIFO; `serve ( -- )` reads one (`rq-line ( -- f )`) and `dispatch ( -- )` picks the answer. The others build answers and exist for the app's protocol only: `answer`, `answer-n`, `refuse`, `rq-word`, `rq-c`, `rq=`, `=s`, `hexnum`, `hexd`, `hex4`, `hex8`, `lf`, `noop`, `on/off`, `dec16` (the length of `store`: decimal, at most 65535), the request words `rq-restart`, `rq-source`, `rq-store`, `rq-autostart`, the payload words `p-echo`, `p-version`, `p-status`, `p-words`, `p-stack`, `p-board`, `p-source`, the helpers `pwm-ctl`, `port-word`, `pwm-word`, `fnx`, `fn2`, `tx-idle`, and the variables and buffers `serving`, `counting`, `acount`, `rq`, `rq#`, `rq>`, `rqn-a`, `rqn-n`, `ea`, `en`, `rlen`, `rclo`, `rchi`, `rflag`, `pp`, `s-in` (the `input` count of a `status` answer, taken once), `dacc`.

6. Examples

Each example can be typed at the terminal line by line, or run from the app's editor. The ones that loop until a key is pressed stop on any key. An input line holds 80 characters, so longer definitions are split over several lines or into smaller words.

Readings shown come from the simulator (chapter 8); on the board you get your own values.

6.1 LEDs and the user button

Count in binary on the LEDs (about 8 seconds)

```
: count256 256 0 do i led! 30 ms loop 0 led! ;
count256
```

Running light

```
: runner begin 8 0 do 1 i lshift led! 100 ms loop key? until key drop 0 led! ;
runner
```

Two LEDs blinking in turn (one LED at a time with `led-on` and `led-toggle`)

```
: alternate ( -- )   \ LED1 and LED2 take turns; any key stops
  0 led! 1 led-on
  begin 1 led-toggle 2 led-toggle 250 ms key? until key drop 0 led! ;
alternate
```

The button lights all LEDs

```
: mirror begin btn@ led! key? until key drop 0 led! ;
mirror
```

`btn@` gives −1 while the button is pressed, which switches all eight LEDs on.

Reaction timer

```
: wait-btn begin btn@ until ;
: wait-free begin btn@ 0= until ;
: react
  ." wait for the LEDs..." cr
  2000 ms 255 led! ms@
  wait-btn ms@ swap - 0 led!
  ." reaction time " . ." ms" cr
  wait-free ;
react
```

Button counter: each press of the USER button counts up, in binary on the LEDs

```
variable presses
: count-press 1 presses +! presses @ led! begin btn@ 0= until ;
: counter
  0 presses !
  begin btn@ if count-press then key? until key drop
  ." pressed " presses ? ." times" cr ;
counter
```

6.2 Digital pins

These need a few parts on the header. Pin numbers are the labels D0..D14 (0..14) and PIO_01..PIO_08 (16..23).

Blink an LED on D3 (LED with a 330 Ω to 1 kΩ resistor from D3 to GND)

```
: blink3 3 output begin 3 toggle 250 ms key? until key drop 3 low ;
blink3
```

A button on D6 (push button from D6 to GND, 10 kΩ from D6 to 3.3 V; with the `-PullUps` build the resistor is not needed). LED1 shows the button.

```
: watch6 begin 6 pin@ 0= 1 and led! key? until key drop 0 led! ;
watch6
```

Loopback: connect D0 to D1 with a wire

```
0 output  0 high  1 pin@ .  0 low  1 pin@ .  0 input
```

prints `-1 0`.

The Pmod header: a counter on PIO_01..PIO_04, through the port register of pins 16..31

```
15 $801C !                  \ PIO_01..PIO_04 as outputs
: pio-count 16 0 do i $801A ! 250 ms loop 0 $801A ! ;
pio-count
```

6.3 Analog inputs, temperature and accelerometer

Voltmeter on AIN3: prints the voltage every half second

```
: volts begin 3 mv@ . ." mV" cr 500 ms key? until key drop ;
volts
```

prints `1650 mV` for 1.65 V, for example.

Bar graph: a potentiometer (ends to 3.3 V and GND, wiper to AIN3) lights 0 to 8 LEDs

```
: level ( -- n ) 3 adc@ 9 4096 */ 1 swap lshift 1- ;
: bar begin level led! 50 ms key? until key drop 0 led! ;
bar
```

In the middle position four LEDs are lit (`level .` prints `15`).

A pin measured by the ADC: connect D3 to AIN0 with a wire

```
3 output  3 high  10 ms  0 mv@ .  3 low  10 ms  0 mv@ .  3 input
```

The first reading is close to 3300 mV, the second close to 0.

Thermometer: one line per second

```
: thermo begin .temp cr 1000 ms key? until key drop ;
thermo
```

prints lines like `34.6 C`.

Tilt: the three axes twice a second

```
: tilt begin accel@ rot . swap . . cr 500 ms key? until key drop ;
tilt
```

prints `0 0 1000` with the board lying flat; tilt it and the 1000 mg move to another axis.

Spirit level: one LED shows the tilt along the x axis

```
: x-led ( -- n ) accel@ 2drop 1000 + 250 / 0 max 7 min ;
: spirit begin 1 x-led lshift led! 50 ms key? until key drop 0 led! ;
spirit
```

Flat, the board lights LED5 (`x-led .` prints `4`); tilted one way or the other, the light moves toward LED1 or LED8.

Which side is up

```
: up? accel@ nip nip 0> if ." face up" else ." face down" then cr ;
up?
```

6.4 PWM and DDS

Dim an LED on D3 (LED and resistor as in 6.2)

```
1000 0 pwm-hz  100 0 pwm-duty  3 0 pwm-pin     \ 10 % brightness
0 pwm-off
```

Breathing LED on D3

```
: ramp-up 100 0 do i 10 * 0 pwm-duty 10 ms loop ;
: ramp-down 100 0 do 1000 i 10 * - 0 pwm-duty 10 ms loop ;
: breathe 1000 0 pwm-hz 3 0 pwm-pin
  begin ramp-up ramp-down key? until key drop 0 pwm-off ;
breathe
```

`+loop` counts upward only, so `ramp-down` computes the falling duty from a rising index.

A tone: a piezo buzzer from D7 to GND

```
440 1 pwm-hz  500 1 pwm-duty  7 1 pwm-pin  500 ms  1 pwm-off
```

PWM measured by the ADC: 10 kΩ from D3 to AIN0, 1 µF from AIN0 to GND

```
10 0 pwm-khz  3 0 pwm-pin
250 0 pwm-duty  100 ms  0 mv@ .
750 0 pwm-duty  100 ms  0 mv@ .
0 pwm-off
```

prints about 825 and 2475 mV: the filter turns the duty cycle into a share of 3.3 V.

A DDS frequency, counted by Forth. The pin's level can be read while the DDS drives it, so Forth can count the edges itself (good up to some tens of kHz):

```
variable epin  variable elast  variable ecount
: edges ( pin ms -- n )   \ level changes on the pin within ms milliseconds
  swap epin ! 0 ecount ! epin @ pin@ elast !
  ms@ + begin
    epin @ pin@ dup elast @ <> if 1 ecount +! then elast !
    dup ms@ - 0<
  until drop ecount @ ;
1000 dds-hz  3 dds-pin  3 1000 edges .  dds-off
```

prints about 2000 (two edges per period; on the board it printed 2001).

A DDS sweep: 100 Hz to 10 kHz in 100 steps on D7 (a piezo buzzer from D7 to GND makes it audible); the phase keeps running from step to step, so the sweep has no gaps

```
: sweep ( -- )
  100 dds-hz 7 dds-pin
  100 0 do i 100 * 100 + dds-hz 20 ms loop dds-off ;
sweep
```

6.5 Serial interfaces

A UART loopback on one pin (nothing to connect: the pin reads back what TX drives)

```
115200. 7 7 0 uart-open                \ UART 0, TX and RX on D7
: send ( -- ) 10 0 do 48 i + 0 uart-emit loop ;
: show ( -- ) begin 0 uart-key? while 0 uart-key emit repeat ;
send 2 ms show
0 uart-close
```

prints `0123456789`. The ten bytes take about a millisecond at 115200 Bd; they wait in the receive buffer until `show` reads them.

A bridge between the terminal and a serial device, for example a GPS module or another board with 3.3 V levels: its TX to D1, its RX to D0, GND to GND

```
: bridge ( -- )   \ terminal <-> UART 0; ESC ends
  begin
    0 uart-key? if 0 uart-key emit then
    key? if key dup 27 = if drop exit then 0 uart-emit then
  again ;
9600. 0 1 0 uart-open bridge
```

Everything the device sends appears in the terminal, and what you type goes to the device.

Eight LEDs on a 74HC595 shift register: SCK (D13) to SH_CP, MOSI (D11) to DS, D10 to ST_CP (the latch), OE to GND, MR to 3.3 V

```
13 11 12 0 spi-open  10 output 10 low   \ MISO (D12) is not used
: >595 ( c -- ) 0 spi-xfer drop 10 high 10 low ;
: count595 ( -- ) 256 0 do i >595 20 ms loop ;
count595
```

counts from 0 to 255 in binary on the eight LEDs. `spi-xfer` shifts the byte out MSB first, the pulse on D10 copies it to the outputs.

An I2C EEPROM (24C02 or similar) at address `$50`, with SCL on PIO3 and SDA on PIO4 (pins 18 and 19) and pull-ups on both

```
100 18 19 i2c-open
i2c-scan                        \ prints 50
65 0 $50 i2c!  5 ms             \ 65 into cell 0; writing takes 5 ms
0 $50 i2c@ .                    \ 65
i2c-close
```

The value stays in the EEPROM over power-off. `i2c-scan` is the first thing to try with a new module: it shows whether the device answers and at which address.

6.6 A buffer in the data RAM

64 readings of AIN3 at 10 ms intervals, kept in the data RAM, then their smallest and largest value:

```
128 buffer: samples
: sample ( i -- a ) cells samples + ;
: record 64 0 do 3 mv@ i sample ! 10 ms loop ;
: span ( -- min max ) 32767 -32768 64 0 do i sample @ tuck max >r min r> loop ;
record span . .
```

prints the largest, then the smallest value (`1650 1650` for a steady 1.65 V). `samples` uses 128 bytes of data RAM and leaves the code space alone: `unused-data .` now prints `7040`.

6.7 Data in the serial flash

The flash keeps data over power-off. This example writes the 64 readings of 6.6 to `$100000` (1 MB into the flash) and reads one back:

```
: f@ ( lo hi -- x ) f-read 0 spi 0 spi f-end 8 lshift or ;
0 $10 f-erase                    \ erase the 4 KB sector at $100000
samples 128 0 $10 f-prog         \ program 128 bytes (within one page)
0 $10 f@ .                       \ the first reading
```

After a power cycle (and `f@` typed in again), `0 $10 f@ .` still prints it. A page holds 256 bytes; for more data, program one page after another (`$100000`, `$100100`, …), and erase each 4 KB sector once before.

A look at the saved source's header

```
: fdump ( lo hi -- ) f-read 16 0 do 0 spi . loop f-end ;
hex 0 0 fdump decimal
```

prints, for example, `48 46 31 36 11 0 0 0 E8 A3 A1 76 0 0 0 0`: the magic `HF16`, the length ($11 = 17 bytes), the CRC and the flags (autostart off).

6.8 The bench demo: examples/bench.fs

`examples/bench.fs` uses a small bench wiring:

Part	Connection
potentiometer	ends to 3.3 V and GND, wiper to AIN3
push button	from D6 to GND, with a 10 kΩ pull-up to 3.3 V (or the `-PullUps` build)
three LEDs	D3, D4 and D5, each with a resistor to GND

```
: pot ( -- mv ) 3 mv@ ;
: bar ( mv -- ) 8 3300 */ 1 swap lshift 1- led! ;
: show ( -- ) ." pot " pot . ." mV  D6 " 6 pin@ . ." user " btn@ . cr ;
: bench-off ( -- ) 0 pwm-off 0 led! 4 low 5 low ;
: bench ( -- )
  4 output 5 output 1000 0 pwm-hz 3 0 pwm-pin
  ms@ begin
    pot dup bar 1000 3300 */ 0 pwm-duty
    6 pin@ 0= 4 pin!
    ms@ over - 499 > if 5 toggle show drop ms@ then
  key? until drop key drop bench-off ;
```

Run it (**Run all** in the app, or `python tools/mforth.py -p COMx -f examples/bench.fs`), then type `bench`. The potentiometer sets the LED bar and the brightness of the LED on D3 (PWM channel 0 at 1 kHz), the LED on D4 lights while the button on D6 is pressed, the one on D5 blinks, and a status line comes every 500 ms:

```
bench pot 2500 mV  D6 -1 user 0
pot 2499 mV  D6 -1 user 0
```

Any key stops it and switches everything off.

6.9 An application that starts with the board

A saved source whose last line starts a loop turns the MAX1000 into a stand-alone device. For example `night.fs`, which lights LED1..LED8 as the input on AIN3 rises:

```
\ night.fs - LED bar from AIN3, runs at power-up
: level ( -- n ) 3 adc@ 9 4096 */ 1 swap lshift 1- ;
: night begin level led! 50 ms key? until key drop 0 led! ;
night
```

Store it with **Store editor** in the app, or with `python tools/mforth.py -p COMx --store night.fs`, and press RESET: the bar runs without a PC. With a terminal connected, any key ends `night` and gives the prompt back, with `level` and `night` defined. Because `night` waits in `ms`, the app can connect while it runs.

The last line is the only one that may run without returning: lines after it would never be read. To change the program, press a key (or restart clean), then store the new version.

7. Differences from standard Forth

HF16 Forth follows the usual Forth conventions (ANS Forth / Forth 2012) where it has the same words, with these differences:

Behaviour

- `'` returns 0 for an unknown word instead of signalling an error, and `0 execute` restarts Forth through `cold`.
- `/`, `mod` and `/mod` use symmetric division, rounding toward zero (the standard allows this or floored division).
- `+loop` counts upward only.
- No warning when a word is redefined.
- Number overflow is not detected; values wrap around at 16 bits.
- The data stack holds at most 20 items at the terminal (section 3.4); the return stack has 32 entries.
- Compile-only words (`if`, `do`, `."`, `[']`, ...) are not checked. Typed outside a definition, they write stray code into the dictionary and leave junk on the stack.
- `buffer:` takes its memory from a separate data RAM (section 5.7) and rounds the size up to even, so buffers stay cell-aligned.
- Code must lie below `$4000`; there is no check for a full dictionary.

Not implemented

- Most double-cell words (`d+`, `d.`, `m*`, `*/mod`, `#`, `<# #>`, ...); only `um*`, `um/mod`, `*/`, `ud/u` and `m+u` exist. Double numbers can be typed with a trailing dot (`115200.`), unsigned only
- `char`, `[char]` and `s"` (the last two exist only in the cross-compiler), `accept`, `evaluate`, `source`
- `?do`, `leave`, `case`, `does>`, `postpone`, `defer`, `marker`, `forget`
- `pick`, `roll`, `2swap`, `2over`, `fill`, `move`, `erase`, `see`, `dump`

Most of these can be added as definitions in `forth/kernel.fs` (chapter 8), or typed in and kept in the saved source.

8. Changing the kernel

The kernel lives in `forth/kernel.fs`, written in Forth. `tools/hf16.py` cross-compiles it on the PC. Because the kernel image is part of the FPGA configuration, a changed kernel needs a new FPGA build. Everything here works in the repository, not in a release.

The cross-compiler understands this subset:

- **Outside definitions:** numbers, `:`, `:inline`, `variable`, `constant`, `create`, `allot`, `,`, `c,`, `immediate`, and comments.
- **Inside definitions:** primitives (always inline), calls to other words (including words defined further down), numbers, `if else then begin until again while repeat do loop +loop i unloop exit recurse ." s" [char] [']`, and `[op] name`, which compiles the opcode of primitive `name` as a number (used by the target compiler).
- `:inline name ... ;` makes a word that the target compiler copies inline.

The kernel and the words typed later share the 16 KB code space, and the kernel's buffers must stay at `$5C00`..`$5FFF` (or move together with `data-end`).

Workflow:

1. Edit `forth/kernel.fs`.
2. Test in the instruction-set simulator. It runs the kernel and feeds it input lines from a file, one Forth line per line (use a file rather than `-i` for lines with double quotes: PowerShell mangles quotes in native arguments): `powershell python tools/hf16.py run forth/kernel.fs -f test.txt python tools/hf16.py run forth/kernel.fs -i "2 3 + ."` The simulator models the LEDs, the pins (inputs read low, except D6, which reads high like the bench button), the ADC (AIN3 at 1.65 V, AIN0 at 3.3 V, the others 0), the temperature code 3640 (about 34.6 °C), the accelerometer (x 0, y 0, z 1000 mg), the serial flash and the CRC unit. Its millisecond counter advances once per 100 instructions, so `ms` runs much faster than on the board. The run ends when all input is read and Forth waits for more (also inside a long `ms` loop), or after `--steps n` instructions. `--flash image.bin` keeps the flash in a file between runs, `--button` holds the USER button. The serial interfaces are modelled per transfer, without timing: `--wire 0-1` joins two pins as a jumper wire would (a UART reaches the UARTs whose RX pin is joined to its TX pin, SPI reads back MOSI when MISO is joined to it), and `--i2c-dev $50` puts a 256-byte register memory on the I2C bus.
3. Run the tests: `sim\run_flash_tests.ps1` (flash words and the saved source), `python sim\test_protocol.py` (the app's requests), `python sim\test_serial.py` (serial interfaces, pin owners, double numbers) and, with Vivado 2023.2 installed, `sim\run_serial.ps1` (the Verilog of the UART, SPI and I2C units against models in a testbench).
4. Build the kernel image and the FPGA configuration. This needs **Quartus Prime Lite 18.1** (with MAX 10 support) and Python 3: `powershell scripts\build.ps1 # kernel + FPGA scripts\build.ps1 -PullUps # the same with weak pull-ups on D0..D14 and PIO1..PIO8` `build.ps1` finds Quartus through `-Quartus <path>`, `QUARTUS_ROOTDIR` or `<drive>:\Quartus\18.1\quartus`. It writes `build\output_files\max1000_forth.sof` and `.pof`.
5. Load it: the `.sof` into the FPGA to try it (until power-off), the `.pof` into the internal flash to keep it (section 1): `powershell quartus_pgm -c 1 -m JTAG -o "p;build/output_files/max1000_forth.sof" quartus_pgm -c 1 -m JTAG -o "p;build/output_files/max1000_forth.pof"`

`tools/hf16.py build forth/kernel.fs -o build/forth` writes the image alone, with `build/forth.map`, the address of every word. The app recognises the board by the name `max1000-forth` in the `version` answer (`p-version`); keep it when you change the version number.

9. Troubleshooting

Symptom	Cause and remedy
No banner, nothing happens	The banner came before you connected: press Ctrl-R (or *Restart* in the app). Check the port (channel B of the board's USB chip) and 115200 baud. If LED1 is dark, MAX1000-FORTH is not running: program it (chapter 1) and press RESET.
`program_flash.ps1` finds no board or no `quartus_pgm`	Install the Arrow USB Blaster driver (the board's built-in programmer) and the Quartus Prime Programmer software, and plug the board in with its USB cable. `quartus_pgm -l` lists what Quartus sees: the board appears as an Arrow USB Blaster.
The app says the board does not answer	Another program has the port, the wrong port is chosen, or the board runs another configuration.
The app's status line shows *no answer* while a word runs	The word only computes: Forth answers the app only at the prompt and in `ms`, `key?`, `key` and output (section 2.3). Wait, or click *Restart*.
The board hangs at power-up, or misbehaves from the start	The saved source does it: hold the USER button while you press RESET, or *Restart clean* in the app (section 4.7). Then switch the autostart off or erase it.
An error ends in `(in the saved source)`	A line of the saved source failed; the lines after it were not run. Fix the source and store it again.
Forth stops responding	A word is in an endless loop or waiting for `key`. Press any key, or Ctrl-R (*Restart* in the app).
Your words are gone	A restart or a power cycle brings back the original kernel. Keep useful definitions in a file and run it, save it as the board's source (section 2.7), or add them to `kernel.fs`.
Pasting a long text into a terminal program loses characters	Forth's receive buffer holds 8 KB. Paste in parts, or use the app's *Run all* or `mforth.py -f`, which wait for Forth.
Numbers look wrong, or every number gives `?`	Check `base`: after `hex` or `2 base !`, input and output both use that base, and an error doesn't reset it. `decimal` switches back.
A long line gives an error near its end	Only 80 characters fit on a line; the rest is dropped, possibly in the middle of a word. Split the line.
A digital input reads at random	It is open: the pins have no pull-ups. Connect it, add a resistor, or build with `-PullUps` (section 5.11).
An output pin doesn't change	The pin is still an input (`n output` first), or a PWM channel, the DDS or a serial interface owns it (`n owner .` tells which; switch it off or close it). Check the pin number against the label (section 5.11).
An analog input shows a voltage with nothing connected	Open inputs show the charge of the channel converted before them (section 5.12).
`1 dds-hz` gives 0.98 Hz	The DDS steps are 0.0447 Hz; at low frequencies the error is a few percent. For an exact value use `dds-tuning` (section 5.14).
`pin taken`	Another PWM channel, the DDS or a serial interface has the pin: `n owner .` tells which (section 5.15). Close it, or pick another pin.

Symptom	Cause and remedy
A UART receives nothing or garbage	TX of one side goes to RX of the other, both sides share GND and use the same baud rate and 8N1; 3.3 V levels only (no RS-232 without a level converter). `uart-status` shows lost bytes (8) and framing errors (16).
`i2c-scan` finds nothing, or `i2c timeout`	SCL and SDA need pull-up resistors to 3.3 V (most modules have them). Check the pins given to `i2c-open` and the module's supply.
SPI reads only 0 or 255	MISO is on the wrong pin or the device is not selected: drive its chip-select pin low around the transfers. Check the mode (`spi-mode`) in the device's data sheet.
A PWM change does not show at once	It waits for the end of the running cycle, so a channel at 1 Hz takes up to a second (section 5.13).
`accel@` returns nonsense inside a flash read	The accelerometer and the flash share the SPI master: end the read with `f-end` first (section 5.17).
Data read back from the flash is wrong	The sector was not erased before programming, or `f-prog` crossed a 256-byte page boundary (section 5.17).
`data RAM full`	`buffer:` asked for more than is left (`unused-data`). A restart frees the data RAM.
The temperature looks a few degrees off	`temp@` is approximate (section 5.12).